

Dynamic Gamma Modulation of Hippocampal Place Cells Predominates Development of Theta Sequences

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Abstract

The experience-dependent spatial cognitive process requires sequential organization of hippocampal neural activities by theta rhythm, which develops to represent highly compressed information for rapid learning. However, how the theta sequences were developed in a finer time scale within theta cycles remains unclear. In this study, we found that sweep-ahead structure of theta sequences developing with exploration was predominantly dependent on a relatively large proportion of FG-cells, i.e. a subset of place cells dominantly phase-locked to fast gamma rhythms. These ensembles integrated compressed spatial information entrained in a theta sequence by cells consistently firing at precessing slow gamma phases within the theta cycle. Accordingly, the sweep-ahead structure of FG-cell sequences was positively correlated with the intensity of slow gamma phase precession, in particular during early sequence development. These findings highlight the dynamic network-modulation by fast and slow gamma in the development of theta sequences which may further facilitate memory encoding and retrieval.

eLife assessment

Using electrophysiological recordings in freely moving rats, this **valuable** study investigates the role of different gamma frequency bands in the development of spatial representations in the hippocampus. However, the evidence is **incomplete** as the methods and data analysis need significant improvement. Critically, alternative interpretations and analyses must be provided, especially regarding the nature of gamma oscillations in the hippocampus and their interaction with neuronal firing dynamics and theta sequence features. This study will be of interest to neuroscientists working in the field of spatial navigation and neuronal dynamics.

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Introduction

The hippocampus is known as its central role in encoding and storing experience and thus is essential for forming episodic memories (Buzsaki, McKenzie, & Davachi, 2022 [↗](#); Cohen & Eichenbaum, 1993 [↗](#)). In the context of active navigation in an environment, hippocampal place cells are organized to fire sequentially within individual cycles of theta oscillations (4-12Hz) and form time-compressed representations of behavioral experiences, referred to as theta sequences (Colgin, 2020 [↗](#); Comrie, Frank, & Kay, 2022 [↗](#); Dragoi & Buzsaki, 2006 [↗](#); Skaggs, McNaughton, Wilson, & Barnes, 1996 [↗](#); Wallis, 2018 [↗](#)). By sweeping ahead from the past to the upcoming locations, theta sequences represent information about potential future paths that could guide goal-directed behaviors (Gupta, van der Meer, Touretzky, & Redish, 2012 [↗](#); Joshi et al., 2023 [↗](#); Zheng, Hwaun, Loza, & Colgin, 2021 [↗](#)). These predictive theta sequences exhibited experience-dependent development when rats were exploring a novel environment or learning a novel goal location (Feng, Silva, & Foster, 2015 [↗](#); Gobbo et al., 2022 [↗](#); Wikenheiser & Redish, 2015 [↗](#); Zheng et al., 2021 [↗](#)). However, how the hippocampal network modulated the development of theta sequences' temporospatial-compressed organization remains incompletely understood.

A common assumption was established by previous studies that theta sequences were manifestations of theta phase precession, in which an individual place cell fired at progressively earlier theta phase as the animal traversed the place field (Aoki et al., 2023 [↗](#); Foster & Wilson, 2007 [↗](#); O'Keefe & Recce, 1993 [↗](#); Qasim, Fried, & Jacobs, 2021 [↗](#); Skaggs et al., 1996 [↗](#)). However, when examined the theta sequences with experience at single trial level, their sequential structures were not developed on the first experience of a novel environment even if the phase precession of individual place cells has been observed (Colgin, 2020 [↗](#); Feng et al., 2015 [↗](#)). This suggests that the dynamic process of theta sequences development could be modulated by another network-level mechanism with neuronal firing organized at a finer timescale within theta cycles. Gamma (25-100Hz) could be a candidate that is thought to interact with theta to temporally organize theta sequences (Dragoi & Buzsaki, 2006 [↗](#); Fernandez-Ruiz, Sirota, Lopes-Dos-Santos, & Dupret, 2023 [↗](#); Lisman & Jensen, 2013 [↗](#)). As two subtypes of gamma, fast and slow gamma rhythms (Colgin et al., 2009 [↗](#); Schomburg et al., 2014 [↗](#); Zhang, Lee, Rozell, & Singer, 2019 [↗](#)) occur at distinct theta phases and coordinate theta sequences in different manners (Zheng, Bieri, Hsiao, & Colgin, 2016 [↗](#)). This indicates that fast and slow gamma may act as different tuners but collaborate together in modulating theta sequence development.

Fast gamma (~65-100Hz), associated with the input from the medial entorhinal cortex, is thought to rapidly encode ongoing novel information in the context (Fernandez-Ruiz et al., 2021 [↗](#); Kemere, Carr, Karlsson, & Frank, 2013 [↗](#); Zheng et al., 2016 [↗](#)). During fast gamma rhythms, place cell spikes occurred across all positions within a place field and displayed theta phase precession (Bieri, Bobbitt, & Colgin, 2014 [↗](#)). This type of cells was identified recently as "phase-precessing" cells, in which the phase-precession pattern emerges with strong fast gamma (Guardamagna, Stella, & Battaglia, 2023 [↗](#)). Their discrete distribution of theta phases is likely to contribute to the sequential structure formation when the sequences have been developed (Wang, Foster, & Pfeiffer, 2020 [↗](#)). However, a question is raised about how these fast gamma-modulated cells be coordinated for the developmental process of theta sequences.

Slow gamma (~25-45Hz) in the hippocampal CA1 reflects information received from CA3, which is responsible for the integration processing of learned information and experience (Colgin et al., 2009 [↗](#); Zhu et al., 2023 [↗](#)). In contrast to fast gamma, spiking during slow gamma exhibited dominant theta phase locking and attenuated theta phase precession (Guardamagna et al., 2023 [↗](#); Zhang et al., 2019 [↗](#)). This concentrated distribution of theta phase could be interpreted as they are coordinated within slow gamma cycles, as a finer time-scale than a theta cycle. Place cells spiking exhibited slow gamma phase precession, to generate mini-sequences in an individual theta

sequence, for highly time-compressed representation of ongoing and future information (Zheng et al., 2016; Zheng et al., 2021). Thus, how fast and slow gamma rhythms coordinate place cells spiking to modulate the dynamic process of theta sequence development remains unclear.

Here, we proposed a scheme that fast gamma rhythms coordinate a subgroup of place cells for rapidly encoding and processing sensory information since from the early phase of theta sequence development (Fig. 1). The developing process would be then dependent on the slow gamma modulation, that the spikes are fired at slow gamma phase-precessed pattern for spatial information compression within theta cycles. To test this hypothesis, we investigated how fast and slow gamma separately modulated place cells activities during the whole process of theta sequence development. We found that the fast gamma phase-locked place cells (FG-cells) were crucial for the development of theta sequences, without which the temporospatially compressed structure of theta sequences could not be developed. Furthermore, these cells exhibited slow gamma phase precession within theta cycles associated with the sweep-ahead structure of theta sequences, which occurred earlier than those cells not phase locked to fast gamma (NFG-cells). Our findings highlight the dynamic network-modulation by neural oscillations fast and slow gamma, at finer timescales within theta, in development of theta sequences which may further facilitate memory encoding and retrieval.

Results

A subgroup of hippocampal place cells phase locked to fast gamma rhythms during active behaviors

To investigate the temporal organization of place cells' firing associated with the gamma rhythms during the development of theta sequences, we recorded neuronal spiking of place cells and local field potentials simultaneously in hippocampal CA1 from 4 freely behaving rats (Fig. 2-S1). As shown in a previous study, 32% of hippocampal pyramidal cells were phase-locked to gamma rhythms during waking theta states (Senior, Huxter, Allen, O'Neill, & Csicsvari, 2008). We further tested how the hippocampal place cells were modulated by slow and fast gamma rhythms respectively. The slow and gamma episodes were detected in time periods when the power of each gamma type (Fig. 2A-B, fast gamma: 65-100 Hz, slow gamma: 25-45 Hz) was above a preset threshold (see Materials and Methods). We then detected the slow or fast gamma phase-locked place cells by quantifying the gamma phase distributions of all spikes across all trials. In this study, 488 place cells were included with place field distributed on the circular track but not located at the resting box. We found that 120 (24.6%) out of these cells were significantly phase-locked to gamma rhythms, a majority of which (113 cells, 23.2%) exhibited significant phase-locking to fast gamma rhythms (Fig. 2C). Their preferred fast gamma phases (202.38 ± 2.01 deg) were concentrated around the peak of fast gamma cycle, which was consistent with the findings in Schomburg et al. 2014 (Schomburg et al., 2014). However, a very small number (3.7%) of cells were found to phase-locked to slow gamma rhythms. Thus, these place cells were subdivided according to whether they were significantly phase-locked to fast gamma (FG-cells) or not (NFG-cells) during active behaviors in the subsequent analyses. Furthermore, the fast gamma phase-locking of FG-cells was stable across trials, with a significantly higher mean vector length of fast gamma phase-locking than those of NFG-cells (Fig. 2D, repeated measure ANOVA, main effect of cell type, $F(1,22)=33.9$, $p=7.5 \times 10^{-6}$, partial $\eta^2=0.6$; main effect of trials, $F(4,88)=1.3$, $p=0.3$, partial $\eta^2=0.06$, no cell type x trials interaction, $F(4,88)=0.5$, $p=0.8$, partial $\eta^2=0.02$). Meanwhile, we also found that a larger proportion of neurons in FG-cells had significant theta phase precession than NFG-cell (Fig. 2-S2, chi-squared test, ratio of theta phase precession cell among the FG-cell vs. NFG-cell, $n=457$ cell, $\chi^2=13.7$, $p=2.1 \times 10^{-4}$, Cramer V=0.2). These results suggested that the FG-cells were stably modulated by fast gamma rhythms during active behaviors.

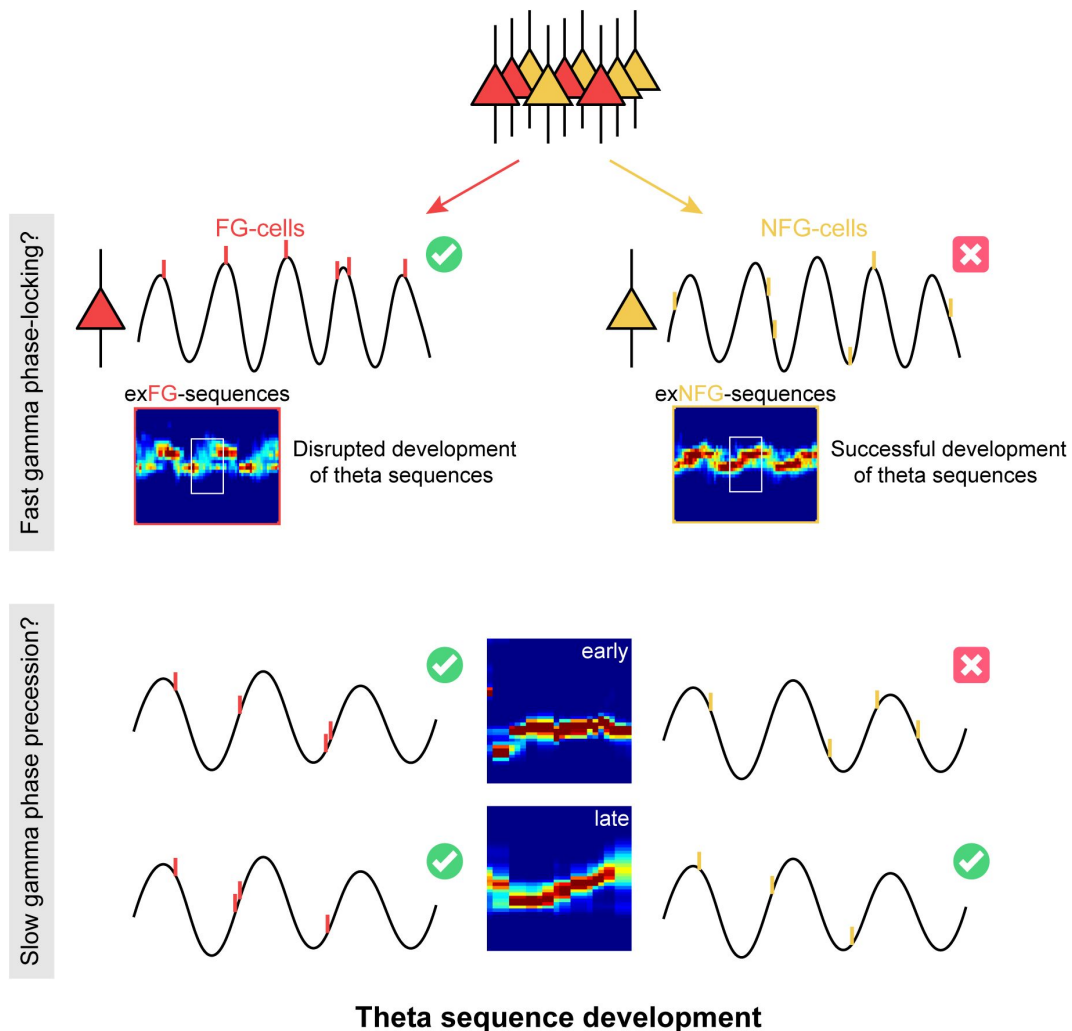


Fig. 1

The schematic of a model for theta sequence development

Fast gamma rhythms coordinate a subgroup of place cells which dominantly fires phase-locked to fast gamma (FG-cells). The development process of theta sequences is disrupted by excluding these FG-cells. A possible model would be that development of sweep-ahead structure of theta sequences is dependent on the slow gamma modulation, i.e. slow gamma phase precession, whereby spatial information could be highly compressed within theta cycles. FG-cells are those cells exhibiting slow gamma phase precession within theta cycles since from the early stage of sequence development, which occurs earlier than the cells not phase locked to fast gamma (NFG-cells).

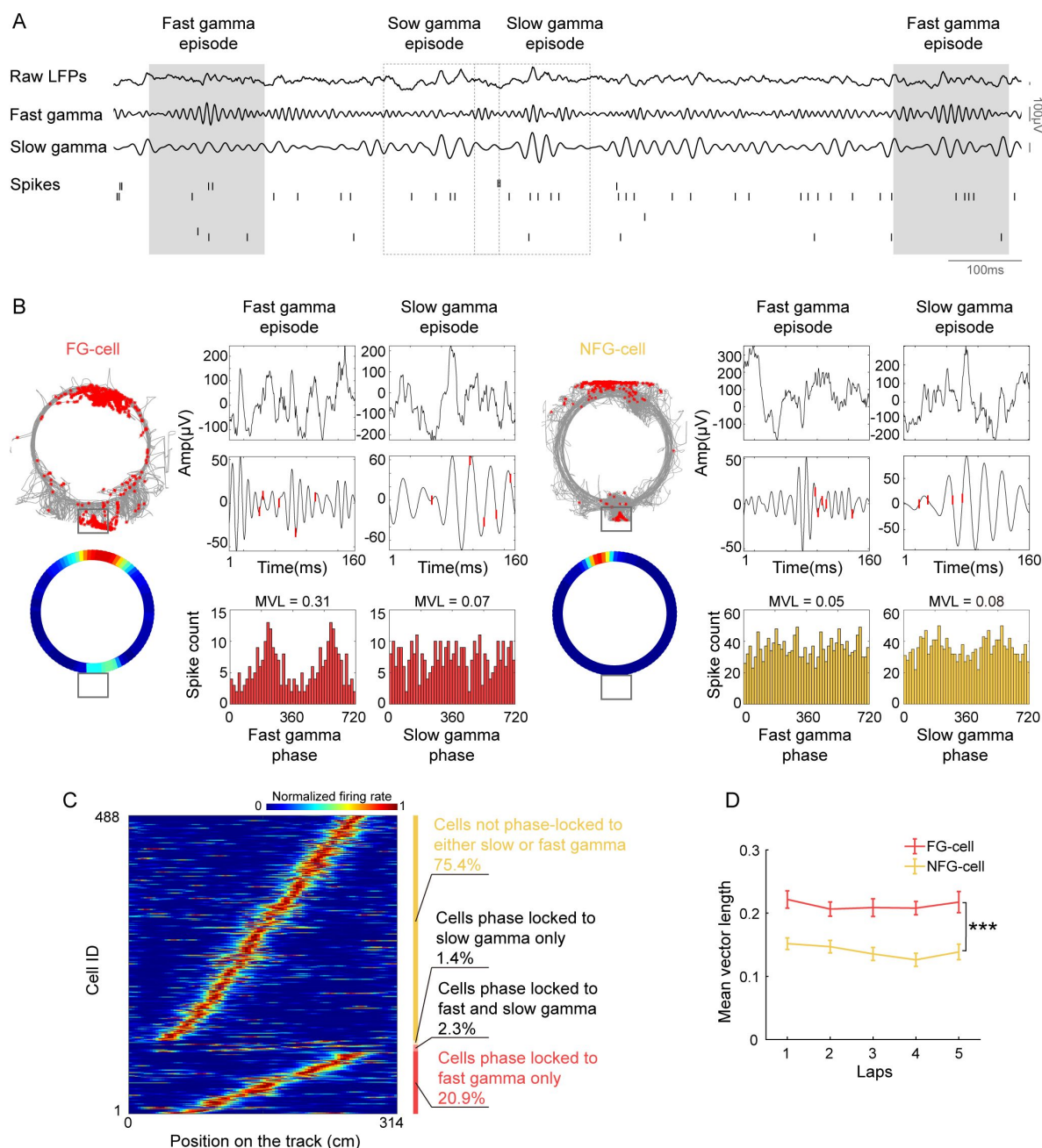


Fig. 2

A subset of hippocampal place cells was modulated by fast gamma rhythms during exploration on a circular track.

(A) An example of simultaneously recorded LFPs (raw LFP, 65-100 Hz 25-45 Hz, bandpass filtered gamma rhythm) and spikes of place cells. (B) Firing characteristics of two examples of FG-cell and NFG-cell, including firing rate map, and phase modulation by fast and slow gamma during detected episodes. Spikes were represented by red lines on top of the gamma waveforms. MVL is short for mean vector length. (C) The normalized turning curve of place cells sorted by the center of mass of their main place field. The proportion of 4 types of cells identified according to different modulation by fast and slow gamma was shown on the right. (D) Mean vector length of FG-cells and NFG-cells across successive running laps ($n=12$ sessions from 4 rats). Data are presented as mean $\pm$ SEM. *** $p<0.001$.

The FG-cells were crucial for theta sequence development

As theta sequences were developed dependent on experience, we next investigated whether FG-cells and NFG-cells played different roles in the development of theta sequences. We then used a Bayesian decoding approach (see Methods) to identify trajectories represented as rats running on a circular track, and detected significant theta sequences in each single lap (**Fig.3A**). Another two decoders were built by excluding either FG-cells spikes or down-sampled NFG-cells spikes respectively, in order to compare the structures of theta sequences without each type of cells. It can be observed that posterior probability distribution of theta sequences was changed by excluding spikes from FG-cells (**Fig.3B**). The weighted correlation of each sequence (Feng et al., 2015) was then measured to quantify the temporospatial correlation of a theta sequence structure. We found that the weighted correlation of theta sequences decoded by excluding FG-cells spikes (exFG-sequences) was much lower than that of theta sequences decoded by excluding down-sampled NFG-cells spikes (exNFG-sequences) in the same theta cycles (**Fig.3C**, repeated measure ANOVA: main effect of cell type, $F(4,44)=128.7$, $p=3.5\times 10^{-48}$, partial $\eta^2=0.05$; post-hoc test, raw sequences vs. exFG-sequences, $p=8.0\times 10^{-56}$, raw sequences vs. exNFG-sequences, $p=1.6\times 10^{-26}$, exFG-sequences vs. exNFG-sequences, $p=4.1\times 10^{-9}$). This suggests that the sequential structure could be dominantly interrupted by excluding FG-cells rather than NFG-cells.

Consistent with the previous studies (Feng et al., 2015; Wang et al., 2020), we also observed the development of theta sequences along with exploration experience (**Fig.3D-E**, repeated measure ANOVA, main effect of trials, $F(4,44)=7.0$, $p=2.0\times 10^{-4}$, partial $\eta^2=0.4$). This could not be accounted for by running speed, as the rats were freely moving at stable running speed across trials (**Fig.3F**, repeated measure ANOVA, main effect of trials: $F(4,44)=0.9$, $p=0.5$ partial $\eta^2=0.07$). We further investigated whether the FG-cells or NFG-cells were required for the development of theta sequence structure. Interestingly, the temporospatial structure of the theta sequences cannot be developed with experience when the FG-cell spikes were excluded, with significantly lower weighted correlation than theta sequences constructed by excluding NFG-cell spikes (**Fig.3D,G**, repeated measure ANOVA, trials x cell type interaction, $F(4,44)=3.0$, $p=0.03$, partial $\eta^2=0.2$, main effect of cell type, $F(1,11)=7.8$, $p=0.02$, partial $\eta^2=0.4$; main effect of trials, $F(4,44)=4.0$, $p=0.008$). Since the theta sequences have been developed after Lap2, we defined the early development phase of sequences as the first 2 laps and late development phase of sequences as the last 2 laps of exploration experience, respectively. We found that the weighted correlation of both exFG-sequences and exNFG-sequences was low during early development (**Fig.3H**, paired t-test, $t(24)=0.2$, $p=0.9$, Cohen's $d=0.04$). During late development, however, the weighted correlation of exNFG-sequences was higher than that of exFG-sequences (**Fig.3H**, paired t-test, $t(24)=-3.7$, $p=0.001$, Cohen's $d=0.8$). These results suggest that fast gamma modulation of place cells may contribute to the development of theta sequence with exploration experience.

In order to estimate to what extent, the FG cells contributed to theta sequence development, we investigated the relationship between the percentage of FG-cells and the formation of sequential structure in different developments. In our dataset, the percentage of FG-cells out of all place cells varied across days, thus we detected theta sequences from trials with relatively low percentage (~10%) and high percentage (~35%) of FG-cells. The number of excluded NFG-cells for decoding was matched up with the number of excluded FG-cells by a pseudo-random method in each day. During early development, the temporospatial structure of theta sequences has not been developed by excluding either low or high percentage of each type of cells (**Fig.4A**). During late development, however, the sweeping-ahead structure of the theta sequence was significantly disrupted when excluding a high percentage (~35%) of FG-cells, which was not observed when excluding a low percentage (~10%) of FG-cells (**Fig.4B**). The sequential structure could be maintained by excluding NFG-cells at either low or high percentage. To quantify the relationship between sequence structure and the number of excluded neurons during early and late development, we then measured the weighted correlation difference between exFG-sequences and

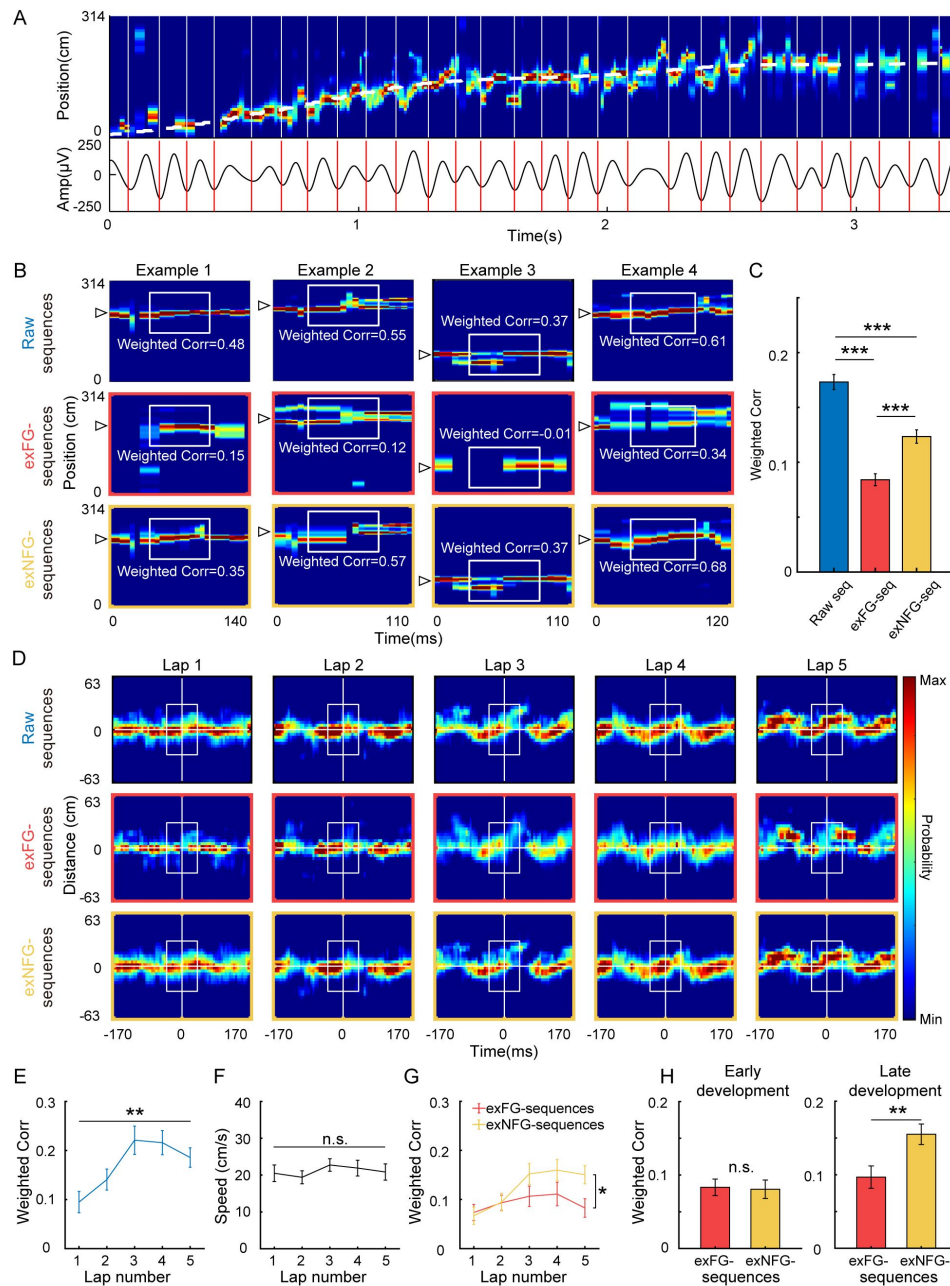


Fig. 3

Theta sequences development was disrupted without FG-cells.

(A) Top, Example color-coded posterior probability distribution across time from a single lap. The running trajectory of the animal was indicated as white dashed line. Bottom, 4–12 Hz bandpass filtered theta rhythm. The beginning and end of theta sequences were indicated as red bars. (B) Examples of detected theta sequences decoded by 3 different decoders, all spikes from all cells (top, i.e. raw sequences), all spikes excluding those from FG-cells (middle, i.e. exFG-sequences) and all spikes excluding those from NFG-cells (bottom, i.e. exNFG-sequences). The white triangle indicates the animal's current position. (C) Weighted correlations between time and decoded positions among three types of sequences. (D) Averaged decoded probabilities by 3 types of decoders over 63 cm centered by the animal's current running position and 170 ms centered by the mid-time point of theta sequence for each lap, in a single recording session. (E) Weighted correlation of raw sequences was significantly increased with running laps ($n=12$ sessions). (F) The running speed did not change across laps. (G) The sweep-ahead structure of the exFG-sequences (red) was disrupted compared to that of exNFG-sequences (yellow). (H) The effect of excluding FG-cells on early (left) and late (right) development of theta sequences. Data are presented as mean $\pm$ SEM. * $p<0.05$, ** $p<0.01$, *** $p<0.001$.

exNFG-sequences as a function of the fraction of excluded cells. The results showed that the weighted correlation difference was close to 0 regardless of the fraction of excluded cells during early development (**Fig.4C** [↗](#), Pearson's correlation, $\text{Corr}=0.1$, $p=0.7$). During late development, however, we found that the weighted correlation difference was positively correlated with the fraction of excluded cells (**Fig.4D** [↗](#), Pearson's correlation, $\text{Corr}=0.6$, $p=0.03$). These results suggest different roles of FG-cells and NFG-cells in the development of the sweeping-ahead structure of theta sequence with experience, that the more temporospatially compressed structure of theta sequences may require a larger percentage of FG-cells.

In addition, we tested whether these results could be accounted for by the different firing characteristics between FG-cells and NFG-cells. We found that although these two types of cells took different proportion of all recorded place cells, the place fields of both types of cells could cover the entire track uniformly (**Fig.2C** [↗](#), **Fig.5A** [↗](#)). We measured the distribution of center of mass (COM) of main place field of FG-cells and number-matched NFG-cells (**Fig.5B** [↗](#)), and found similar distribution of place field COM between cell types (**Fig.5C** [↗](#), two-sample Kolmogorov-Smirnov test, $Z=0.1$, $p=0.8$). Furthermore, we found that the theta sequences in the above analyses were decoded by excluding similar numbers of spikes from FG-cells and NFG-cells (**Fig.5D-E** [↗](#), Student's t test, spike number: $t(22)=1.8$, $p=0.09$, Cohen's $D=0.7$; mean firing rate in field: $t(22)=2.0$, $p=0.06$, Cohen's $D=0.8$), indicating that the different development patterns between exFG-sequences and exNFG-sequences were not due to quality difference between two decoders. The two type of cells exhibited similar place fields size (**Fig.5F** [↗](#), $t(22)=0.7$, $p=0.5$, Cohen's $D=0.3$) and spatial information (**Fig.5G** [↗](#), $t(22)=1.0$, $p=0.3$, Cohen's $D=0.4$). These findings support the hypothesis that FG-cells, but not NFG-cells, may contribute to the development of theta sequences with experience.

FG-cells exhibited slow gamma phase precession for information compression required in theta sequence development

In our previous study, we found that the place cell spikes occurred at progressively earlier slow gamma phases across slow gamma cycles in theta sequences, so-called slow gamma phase precession (Zheng et al., 2016 [↗](#)). As sequences of locations were represented at different slow gamma phases, theta sequences exhibited temporospatially compressed structure representing long paths. We next investigated if the development of theta sequence structure was related to the slow gamma phase modulation of place cells. By detecting the slow gamma phases of spikes within all theta cycles, we observed that the slow gamma phases significantly shifted across successful slow gamma cycles (**Fig.6A** [↗](#), multi-sample Mardia-Watson-Wheeler test, $W(4)=369.4$, $p<2.2\times10^{-16}$; post-hoc test, Cycle -1 vs. Cycle 0, $W(2)=97.5$, $p<2.2\times10^{-16}$, Cycle 0 vs. Cycle 1, $W(2)=95.4$, $p<2.2\times10^{-16}$), which was consistent with the previous finding (Zheng et al., 2016 [↗](#)). Then we separated out the spike activity from the early and late development, and tested if the slow gamma phase precession still existed during different developments. Interestingly, we observed slow gamma phase shifting, but not phase precession, across successive gamma cycles during early development. The distribution of slow gamma phases in Cycle0 was similar as that in Cycle-1, then was dispersed in Cycle1 (**Fig.6B** [↗](#), multi-sample Mardia-Watson-Wheeler test, $W(4)=18.5$, $p=9.8\times10^{-4}$; post-hoc test, Cycle -1 vs. Cycle 0, $W(2)=0.8$, $p=0.7$; Cycle 0 vs. Cycle 1, $W(2)=15.5$, $p=4.4\times10^{-4}$). However, the slow gamma phase precession existed during late development (**Fig.6C** [↗](#), multi-sample Mardia-Watson-Wheeler test, $W(4)=21.2$, $p=2.8\times10^{-4}$; post-hoc test, Cycle -1 vs Cycle 0, $W(2)=8.8$, $p=0.01$; Cycle 0 vs. Cycle 1, $W(2)=8.2$, $p=0.02$). This finding suggests that slow gamma phase precession of place cells may occur with exploring experience.

Furthermore, we investigated how the slow gamma phase precession occurred in theta sequences developed with experience. We detected theta sequences during slow gamma episodes and measured the distribution of phase shifts across gamma cycles within single unites. In the significant theta sequences during slow gamma episodes, we observed the slow gamma phase

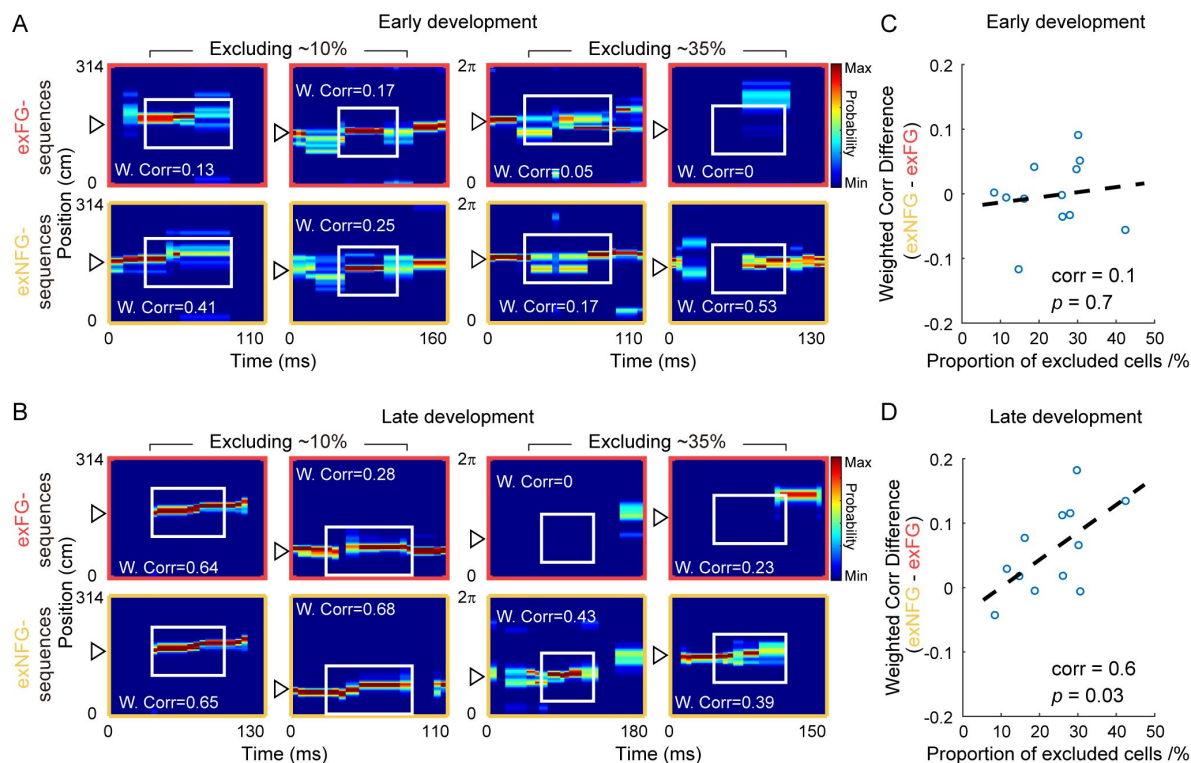


Fig. 4

Temporospatially compressed structure of theta sequences required a relatively large proportion of FG-cells.

(A) Examples of detected theta sequences in early development. Each column is a pair of theta sequences decoded by 2 different decoders, exFG-decoder (top, all spikes excluding those from FG-cells) and exNFG-decoder (bottom, all spikes excluding those from NFG-cells), in a same theta cycle. The left 2 columns of sequences were detected from laps with a relatively small proportion (~10%) of FG-cells. The right 2 columns of sequences were detected from laps with a relatively large proportion (~35%) of FG-cells. W.Corr was short for weighted correlation of each sequence. (B) Same as (A), but for late development of theta sequences. (C) The difference of weighted correlation of sequences between exNFG-decoder and exFG-decoder in early development, as a function of the proportion of excluded cells. Each scatter represents data from each recording session. The dashed line is the linear regression line of all scatters. (D) Same as (C), but for late development of theta sequences. The difference of weighted correlation of sequences between exNFG-decoder and exFG-decoder in late development, was positively correlated with the proportion of excluded cells.

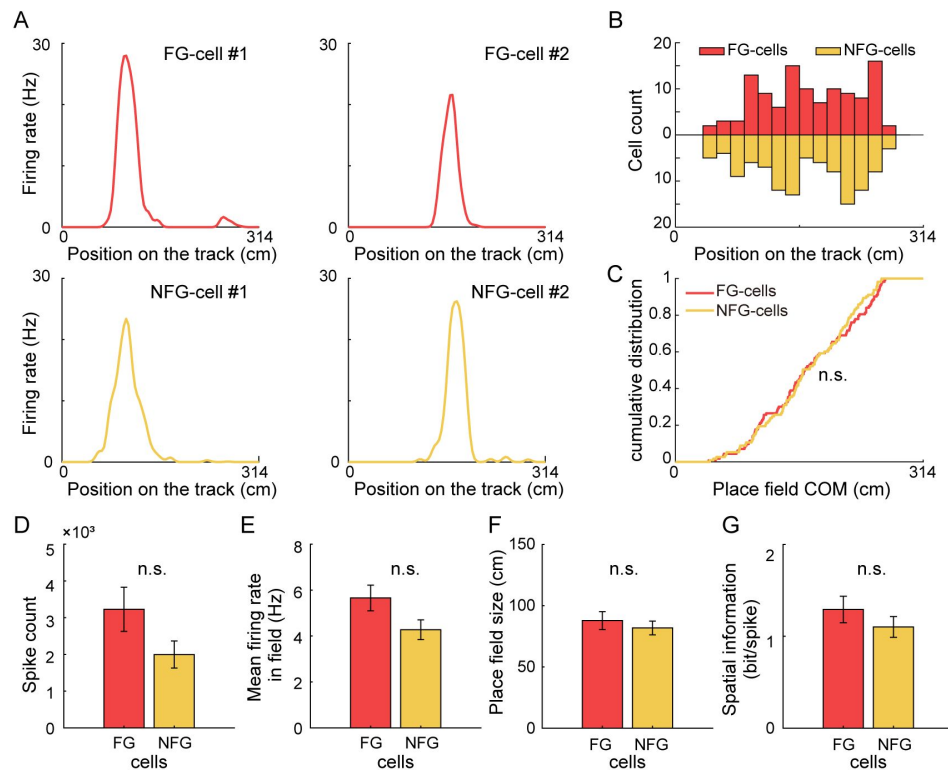


Fig. 5

FG-cells and NFG-cells exhibited similar firing characteristics.

(A) Spatial tuning curves of representative FG- and NFG-cells during exploration on the circular track. Each column of cells exhibited similar place field positions on the track. (B) Distribution of the place field centers of mass (COM) of FG-cells and NFG-cells as a function of position on the track. (C) Cumulative distribution of FG-cells and NFG-cells' place field COM. (D) Spike counts, (E) Mean firing rate in the place field, (F) Place field size, and (G) Spatial information FG-cells and NFG-cells. Data are presented as mean $\pm$ SEM.

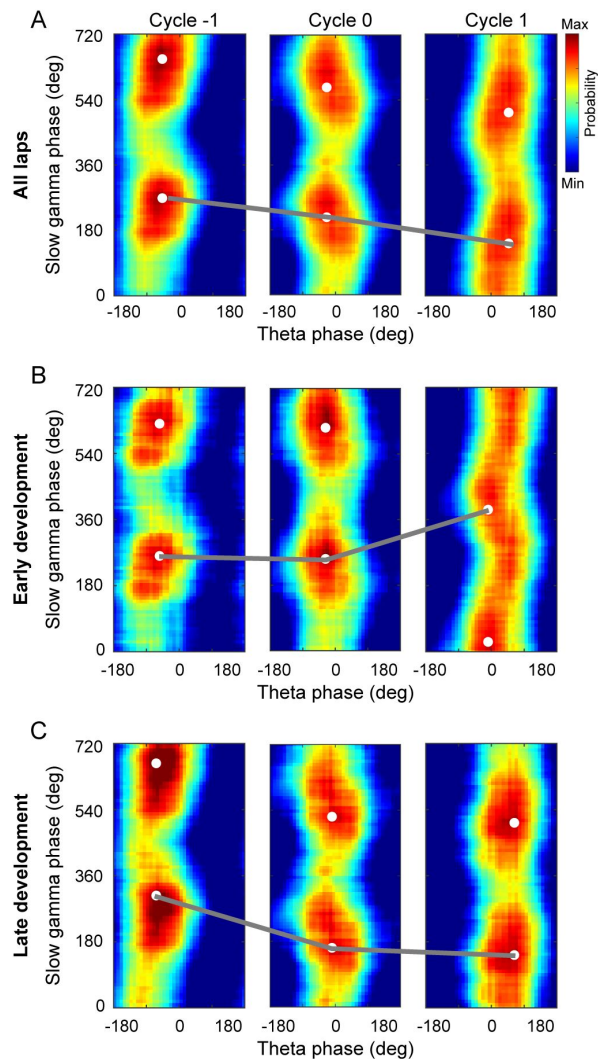


Fig. 6

Slow gamma phase precession across successive gamma cycles occurred during late development of theta sequences.

(A) Probability distributions of slow gamma phases of spikes across successive slow gamma cycles within theta sequences. Gamma cycles within theta cycles were ordered, centered at cycle 0 (i.e., gamma cycle with maximal spiking). Slow gamma phases of spikes shifted systematically across successive slow gamma cycles. (B) Spikes' slow gamma phase did not significantly backward-shifted across successive slow gamma cycles during early development of theta sequences. (C) Same as (A), spikes' slow gamma phase significantly backward-shifted across successive slow gamma cycles during late development of theta sequences. The white dots denote the peak probability of the histogram.

precession from single unit, that the place cell spikes occurred at earlier slow gamma phases in later slow gamma cycles (**Fig.7A**). To quantify the slow gamma phase precession at single unit level, we calculated the phase shift as the phase difference within spike pairs from adjacent active slow gamma cycles (See Methods). We found that slow gamma phase difference was significantly lower than 0 between 2 active slow gamma cycles (**Fig.7B**, one sample t-test, $t(485)=-11.9$, $p=6.3\times 10^{-29}$, Cohen's $D=0.5$), as well as among 3 or ≥ 4 active slow gamma cycles for strict limitation of theta sequences modulated by slow gamma (3 active cycles, $t(140)=-6.8$, $p=3.0\times 10^{-10}$, Cohen's $D=0.6$; ≥ 4 active cycles, $t(27)=-3.2$, $p=0.003$, Cohen's $D=0.6$). Furthermore, we averaged the slow gamma phase differences across different cells within a theta sequence and compare their distribution along with learning trials. In the first lap, we found that the slow gamma phase differences were not significantly different from their surrogate data (**Fig.7C**, median of phase difference from lap1 was in 95% confidence intervals of those from shuffled data, $p=0.3$). However, the distribution of slow gamma phase difference was negatively biased compared with their surrogate data (**Fig.7C**, median of phase difference from lap2-5 was beyond the 95% confidence intervals of those from shuffled data; lap2 $p=0.001$, lap3 $p=0.001$, lap4 $p=0.002$, lap5 $p=0.002$). These results can also be replicated by averaging the slow gamma phase differences from a single unit across successive theta sequences (**Fig.7D**, lap1 $p=0.1$, lap2 $p=0.001$, lap3 $p=0.001$, lap4 $p=0.002$, lap5 $p=0.01$).

These findings raised a question of whether the development of slow gamma phase precession existed for both FG-cells and NFG-cells. To address this question, we quantified the distribution of slow gamma phase difference in FG-cells and NFG-cells respectively. We found that the slow gamma phase difference of FG-cells consistently exhibited a stable phase precession across trials, independent with learning experience (**Fig.7E**, lap1 $p=0.007$, lap2 $p=0.006$, lap3 $p=0.007$, lap4 $p=0.002$, lap5 $p=0.02$). In contrast, the slow gamma phase precession of NFG-cells was experience-dependent, with negative biased distribution of phase difference occurring from Lap2 (**Fig.7E**, lap1 $p=0.9$, lap2 $p=0.01$, lap3 $p=0.001$, lap4 $p=0.009$, lap5 $p=0.1$). These results suggest that FG-cells and NFG-cells have different dynamic modulated by slow gamma rhythms during exploration experience. The slow gamma modulation of FG-cells may occur earlier than that of NFG-cells, thereby driving the location information to be coded within theta cycles in a highly temporospatially compressed way.

FG-cells and NFG-cells exhibited different dynamics of slow gamma modulation along with theta sequence development

Finally, we wondered if there was correlation between slow gamma phase precession and the development of theta sequence structure, and how the FG-cells and NFG-cells coordinated during this process. Within each theta sequence, we measured its weighted correlation of the sequence structure, and plot it as a function of slow gamma phase difference across slow gamma cycles. Interestingly, we found most data were distributed in the Quadrant1 (Q1) compared to the shuffled data distributed at the center (**Fig.8A-B**). The theta sequences with negative slow gamma phase difference and positive weighted correlation were significantly dominant (**Fig.8C**, the probability of observed data from lap1 exceeded 95% confidence intervals of those from shuffled data, $p=0.001$). This finding supports our hypothesis that the theta sequence development could be associated with the slow gamma phase precession.

We further investigated the contribution of FG-cells and NFG-cells to the slow gamma modulated theta sequence development during different exploring phases. Importantly, the distribution of FG-cell-dominant theta sequences was significantly biased to Q1 than that of the shuffled data, not only in late development when the theta sequences have been developed, but also in early development (**Fig.8D-F**, early development, $p=0.001$; late development, $p=0.004$). However, for NFG-cell-dominant theta sequences, their distribution was significantly biased to Q1 than that of the shuffled data only in late development (**Fig.8G-I** bottom, $p=0.046$), but not in early development (**Fig.8G-I** top, $p=0.08$). Taken together, these results support the hypothesis that FG-

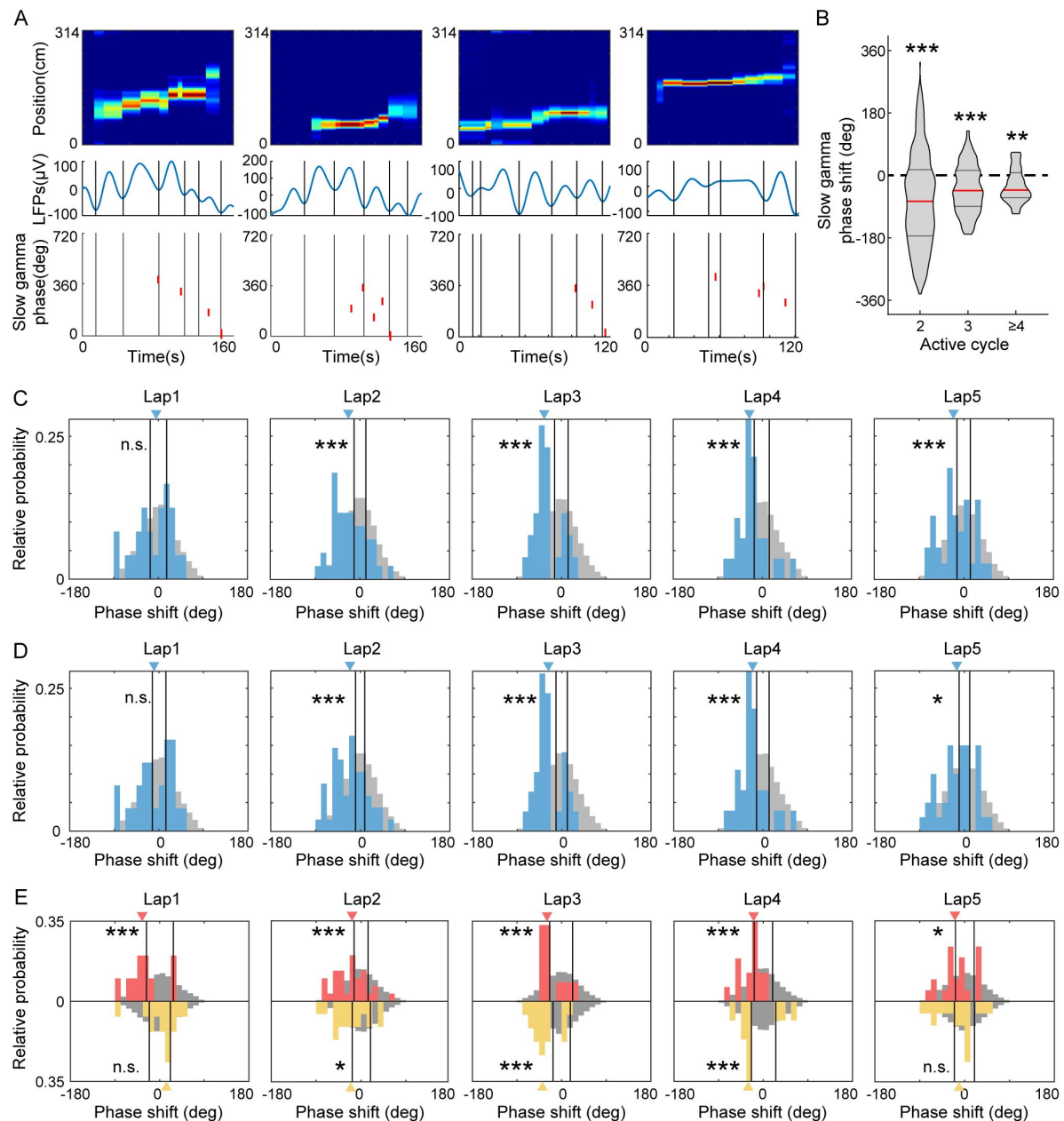


Fig. 7

FG-cells constantly exhibited slow gamma phase precession across laps.

(A) Examples of slow gamma phase precession within theta cycles on single cell level. The top row shows posterior probability for theta sequences. The middle row shows the LFPs waveform in the same theta cycle of the theta sequence, with each slow gamma cycle divided by black lines. The bottom row shows the slow gamma phases of spikes from a representative cell activated in this theta cycle. The spikes occurred at earlier slow gamma phase in the late slow gamma cycles compared with those in the early slow gamma cycles. (B) The averaged slow gamma phase shift across N active cycles were significantly negative ($N=2, 3$ or ≥ 4). The red lines indicate the median and the black lines indicate the 25% and 75% quantile of group data. (C) Histogram of mean slow gamma phase shift averaged within each theta cycle of each lap. The histograms of the real data are shown in blue with blue triangle indicating median. The histograms of the mock data are shown in grey with black lines indicating the 95th confident interval of the their median. (D) The same as (C) but for histogram of mean slow gamma phase shift averaged within each place cell of each lap. (E) Histogram of mean slow gamma phase shift averaged within each FG-cell (above x-axis) or NFG-cell (below x-axis) of each lap. The red and yellow triangles are the median of phase shift in FG-cells and NFG-cells respectively. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

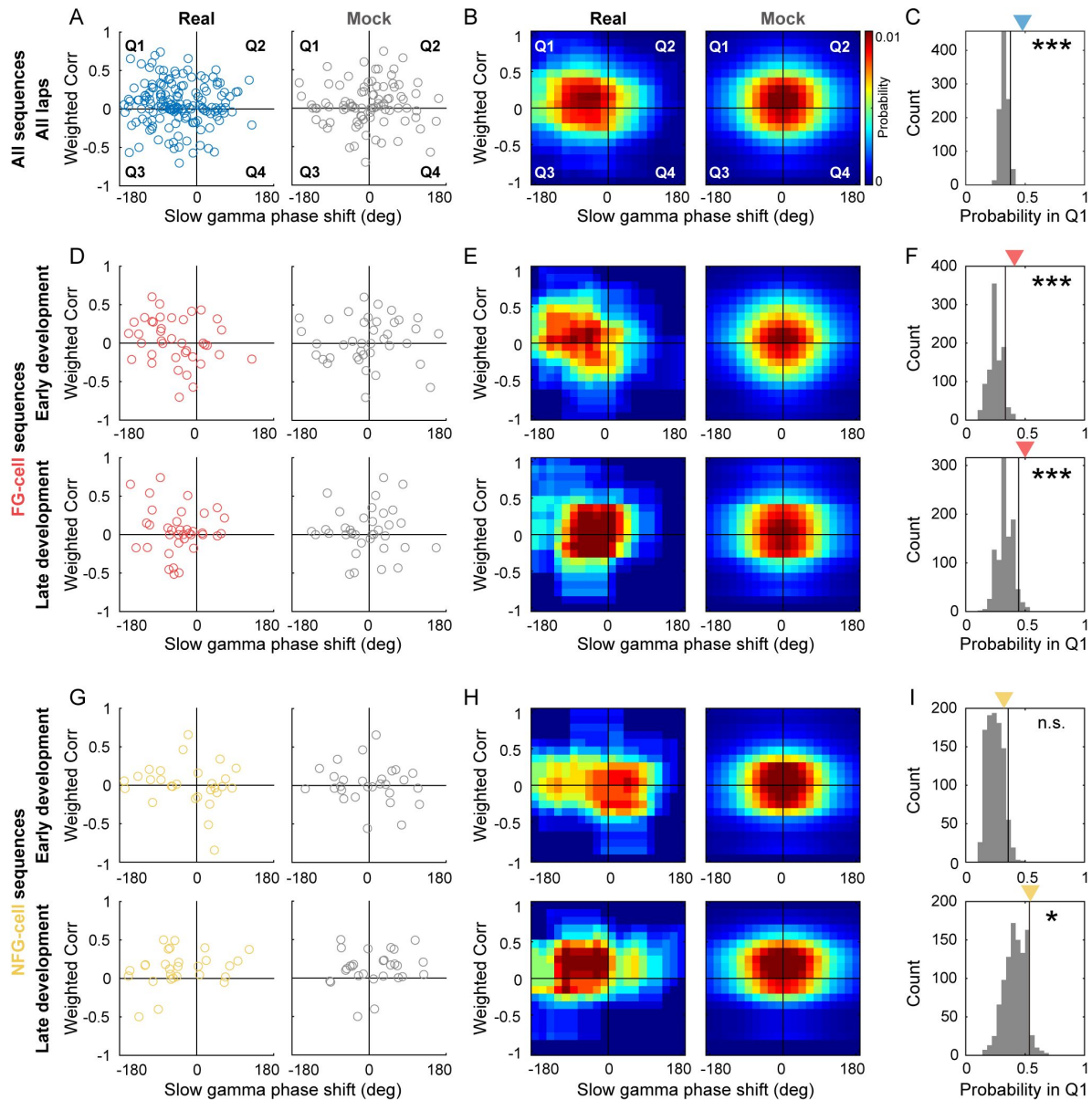


Fig. 8

The sweep-ahead structure of FG-cell sequences was positively correlated with the intensity of slow gamma phase precession during both early and late sequence development.

(A) Scatter plot of averaged slow gamma phase shift across slow gamma cycles within a theta cycle as a function of weighted correlation of all sequences during all laps. Left panel shows real data (blue) and right panel shows mock data by shuffling (gray). (B) Probability distributions of sequences falling in 4 quadrants (Q1-Q4) with slow gamma phase shift (<0 or >0) and weighted correlation (<0 or >0). (C) The count of sequences falling in Q1 (blue triangle) was significantly higher than 95% quantile of the shuffled data (black line). The gray histogram indicates the relative probability of mock data falling in Q1 by 1000 times shuffling. (D) Scatter plot of averaged slow gamma phase shift across slow gamma cycles within a theta cycle as a function of weighted correlation of FG-cell sequences during early development (top) and late development (bottom) of sequences. (E) Probability distributions of FG-cell sequences falling in 4 quadrants during early development (top) and late development (bottom) of sequences. (F) The count of FG-cell sequences falling in Q1 (red triangle) was significantly higher than 95% quantile of the shuffled data (black line) during both early development (top) and late development (bottom) of sequences. (G-H) Same as (D-E), but for NFG-cell sequences. (I) The count of NFG-cell sequences falling in Q1 (red triangle) was significantly higher than 95% quantile of the shuffled data (black line) only during late (bottom) but not early (top) development of sequences. * $p<0.05$, *** $p<0.001$.

cells and NFG-cells may exhibit different dynamical characteristics in gamma modulation, thus their coordination may contribute to the development of theta sequences. FG-cells were consistently modulated by slow gamma phase precession, which would be precondition for theta sequence development.

Discussion

In current study, we employed a circular track exploration task to investigate the different roles of fast and slow gamma rhythms in the coordination of neurons during the experience-dependent development of theta sequences. We first found that a subgroup of place cells (FG-cells) that were phase-locked to fast gamma played a crucial role in the development of theta sequences. By excluding a sufficiently high proportion of FG-cells, the sweep-ahead structure of theta sequences was significantly disturbed compared to that of excluding equivalent number of NFG-cells. These FG-cells exhibited slow gamma phase precession within single theta cycles throughout the developing process of theta sequences. Finally, we showed positive correlation between the intensity of slow gamma phase precession and development of the sweep-ahead structure of theta sequences.

Fast gamma episodes were detected with dominantly high probability as initial exploration of each day and decreased probability as being familiar to the environment (Bieri et al., 2014). This could be related to the feedforward communication from upstream to downstream regions by using fast gamma frequencies (Vinck et al., 2023). Although the information propagation of visual stimulation was unlikely synchronized between visual cortex and hippocampal CA1 (Schneider, Tzanou, Uran, & Vinck, 2023), the MEC inputs could also be entrained by fast gamma to process ongoing sensory information to the hippocampus (Buzsaki & Schomburg, 2015; Colgin et al., 2009). Thus, the fast gamma phase-locked ensemble may act as a receiver of novel information that organized its intrinsic individual neurons in a time-compressed and spatially-ordered sequence to integrate the information under the mesoscopic cellular substrate modulated by fast gamma rhythms (Fernandez-Ruiz et al., 2023). These neurons exhibited similar firing characteristics as those of fast gamma non-phase-locked neurons, however, they showed higher probability presenting theta phase precession. Therefore, this fast gamma phase-locked ensemble likely has overlapping with “bimodal cells” involved in the formation of theta sequence structure including reverse and forward windows (Chu et al., 2023; Wang et al., 2020). In this case, regardless of these neurons, the sweep-ahead structure of theta sequences could not be developed with learning, with the extent of structure impairment positively correlated with the proportion of excluded fast gamma phase-locked neurons.

The important role of fast gamma in modulating theta sequence development was also supported by a specific perturbation experiment. Liu and colleagues optogenetically entrained the MEC GABAergic cells at an artificial gamma frequency (53Hz) and observed the loss of phase precession patterns in CA1 place cells, and thereby the disrupted development of theta sequences (Liu, Todorova, Tang, Oliva, & Fernandez-Ruiz, 2023). The failure of sequential firing organization during behaviors impaired the formation of a predictive map related to the highly time-compressed awake replay sequences executing information storage. Fast gamma was also found diminished between MEC and CA1 in transgenic model mice of Alzheimer’s Disease, linked with spatial memory impairment in AD as a circuit mechanism (Jun et al., 2020). On the other hand, our previous study showed that fast gamma-dominated theta sequences represented shorter trajectories with relative less spatial information within a theta cycle (Zheng et al., 2016). This is indeed not contradictory with the current results, as we found that slow gamma, related to information compression, was also required to modulate fast gamma phase-locked cells during sequence development. We replicated the results of slow gamma phase precession at ensemble level (Zheng et al., 2016), and furthermore observed it at late development, but not early

development, of theta sequences. Thus, a scenario could be drawn out that the network modulator was transitioned from fast gamma to slow gamma throughout the sequence development with learning.

At single neuron level, we also observed that spikes of an individual place cell occurred at precessed slow gamma phases across consecutive slow gamma cycles under the time-scale of a theta window. At a finer scale within a theta cycle, slow gamma cycles coordinated place cells in a “mini-sequence” way (Zheng et al., 2016 [\[link\]](#)), in order to integrate information of long spatial trajectory related to the memory retrieval and predictive behaviors. This negatively-biased phase difference distribution appeared from lap2, when the sweep-ahead structures of theta sequences haven’t been fully developed, suggesting that slow gamma phase precession of place cells contributed to the information compression required to the theta sequence development. More interestingly, we found that slow gamma phase precession of a subset of neurons was absent at initial exploration when the theta phase precession has already displayed (Feng et al., 2015 [\[link\]](#)). This finding was in line with a previous study showing that silencing of CA3 output didn’t affect theta phase precession of a single cell, but impaired theta sequence structures (Middleton & McHugh, 2016 [\[link\]](#)). A possibility was that recruitment of PV-interneurons during the CA3-driven descending phase of the theta cycle, associated with the induction of slow gamma oscillations in CA1, was found reduced in the absence of CA3 input (Fernandez-Ruiz et al., 2023 [\[link\]](#); Topolnik & Tamboli, 2022 [\[link\]](#)). We also provided additional evidence about the relationship between slow gamma phase precession and theta sequence formation, that the more negative slow gamma phase difference across cycles, the more sweep-ahead structure represented at late development of theta sequences.

It is noteworthy that the subset of neurons (FG-cells) was modulated by both fast gamma and slow gamma rhythms during the theta sequence development (**Fig.1** [\[link\]](#)). These place cells, associated with both memory encoding and retrieval, seem to function like memory engram-like representations (Goode, Tanaka, Sahay, & McHugh, 2020 [\[link\]](#); Josselyn & Tonegawa, 2020 [\[link\]](#); Lamothe-Molina et al., 2022 [\[link\]](#); Pettit, Yap, Greenberg, & Harvey, 2022 [\[link\]](#)). Indeed, the proportion of c-Fos positive place cells (putative engram cells) had higher proportion being phase-locked to fast gamma than the c-Fos negative place cells (Tanaka et al., 2018 [\[link\]](#)). In contrast, the NFG-cells exhibited experience-dependent phase coding, that they likely contributed to the path representation at the late learning process. Thus, further studies would be needed to investigate how memory engram cells contribute to theta sequence development and facilitate spatial memory and navigation.

Materials and methods

Subjects

Four male Long-Evans rats weighing 500 to 700g were used in this study. Rats were housed in custom-made acrylic cages (40cm×40cm×40cm) and exposed to a light cycle opposite to natural light (light exposure time: 8:00pm-8:00am). All waking activity recordings were performed in the dark (recording time: 8:00am-8:00pm). Rats were handled at the time of electrode implantation surgery and underwent at least 7 days of behavioral pretraining. Behavioral training was initiated after at least one week of recovery from implantation surgery. Rats were physically deprived during experimental data collection so that their body weight was maintained at ~90% of their free-feeding body weight. All experiments were conducted according to the guidelines of the Animal Care and Use Committee of Tianjin University.

Behavioral paradigm

We designed a circular track apparatus for this study. The circular track with an inner diameter of 90 cm, height of 50 cm, and width of 10 cm, was placing an acrylic resting box (height of 31 cm, length of 28 cm, and width of 21cm) as beginning and ending. Each rat underwent training to navigate on a circular track. A total of eighteen small black silicone dots were utilized as context cues along the circular track, covering approximately two-thirds of its length. Each dot was at a distance of 12 cm (or equivalently 0.28 radians) away from its neighbors. Before surgery, all rats were pre-trained on the same circular track for 1 week and recovered from surgery for at least one week before behavioral training resumed. For every recording session, rats ran 5 trials (or laps) on the circular track. For each running lap, the rats started from the rest box, navigating clockwise to the rest box, and then returned to the rest box counter-clockwise without receiving any food reward. A tone (1kHz sine wave with a duration of 0.25s) was delivered at the beginning and end of the trial. According to the recorded position, the 2D movement speed of the rat was calculated by 2D coordinates and elapsed times of consecutive two points.

Surgery and tetrode implanting

Before surgery, tetrodes were built from 17 μ m polyimide-coated platinum-iridium (10/90%) wire (California Fine Wire, Grover Beach, California). The tetrode tip was platinum-plated using a NanoZ system (White Matter, Seattle, Washington), ensuring that the impedance of each recording channel was below 300 kOhms to facilitate single-neuron activity recording. Eighteen independently mobile tetrodes hyperdrive were surgically implanted into the hippocampal dorsal CA1 region. The rat brain stereotaxic coordinates of the center of the electrode bundle were anterior-posterior [AP] -3.5 mm, medial-lateral [ML] 3.0mm and on day of implantation the tetrode tips moved 1mm downward (dorsal-ventral [DV] 1mm). Screws were placed on the surface of the skull. The hyperdrive bundle were attached to the screws and skull with dental cement. Two screws placed in the skull above the cerebellum served as a panel ground for recording. After surgery, tetrodes were gradually downward to dCA1 stratum pyramidale. One tetrode was designed as recording reference. Recording reference was used to differentiate from other recording tetrode to monitor signal During other recording tetrode downwards, the recording reference was moved up and down to a quiet position in the cortex above the hippocampus and remained quiet throughout signal recording session.

Data acquisition

Behavioral and electrophysiological data were acquired using a Digital Lynx system and Cheetah recording software (Neuralynx, Bozeman, Montana). LFPs recorded continuously from a single channel in each tetrode, within the frequency range of 0.1-500 Hz from raw signal at 2,000 Hz sampling rate. Before each recording session starting, the input amplitude ranges were carefully adjusted to optimize resolution and prevent signal saturation. Typically, the input ranges for LFPs were within the range of $\pm 2,000 \mu$ V to 3000 μ V. To detect unit activity, signals from each channel in a tetrode were bandpass filtered from 600 to 6,000 Hz. The spike detected threshold was 55 μ V. All the recorded signals were obtained by differentiating with the reference tetrode (see “surgery and tetrode planting”), eliminating a large amount of background noise. The position of the rat during navigation behavior was recorded by the video tracking function within Neuralynx system. The video resolution of 720 $\times$ 480 pixels and a frame rate of 25 frames per second. The actual position was tracked with a red light-emitting-diode (LED) on one of the three HS-27-Mini-LED headstage (Neuralynx, Bozeman, Montana) attached to the hyperdrive. The actual 2D coordinates were obtained according to the pixel value where the red is brightest

Spike sorting

Detected spikes were manually sorted using graphical cluster-cutting software (MClust; A.D. Redish, University of Minnesota, Minneapolis). Spikes were sorted using two-dimensional projections of three different features of spike waveforms (i.e., energies, peaks, and peak-to-valley differences) from four channels. Clusters were judged based on waveform shape, isolation distance, and lack of inter-spike-interval violation. Units with mean firing rates of more than 15 Hz were identified as putative fast-spiking interneurons and were excluded from further analysis. The units with peak firing rates more than 1 Hz but mean firing rates of less than 15 Hz was putative pyramidal cell for further analysis.

Spatial firing maps and place cell identification

The animal position on the circular track was linearized to increasing radian values and divided into 90 bins (approximately 0.035 rad per bin and that is 3.5cm). Spatial firing maps were calculated only during locomotor periods (> 5cm/s). To calculate directional spatial firing rate-maps, the number of directional spikes was divided by the time of occupancy at each position bin on the circular track with the same direction, followed by smoothing with a Gaussian kernel (standard deviation of 2 bins). For identifying the place cell, we used all trial spikes to calculate rate-maps. A continuous span of at least three consecutive bins in the rate-map with a peak firing rate exceeding 1 Hz was calculated as a place field. A pyramidal cell contained at least one place field was defined as a place cell. The place field size was defined by 0.1Hz firing rate boundary. The firing rate in each place field is divided by the number of bins in the field to calculate the mean firing rate in the field, the peak firing rate. The primary place field used for analysis was defined as the field with the maximal peak firing rate among all fields. Finally we accepted 488 place cells with place field distributed on the circular track (region from dot 1 to 18) but not located at the resting box.

Spatial specificity was quantified by computing the spatial information score (Langston, Ainge et al. 2010). For each cell, the spatial information score in bits per spike was calculated from the recording in each period as:

$$\text{Spatial information} = \sum_i p_i \frac{\lambda_i}{\lambda} \log_2 \frac{\lambda_i}{\lambda}$$

Where λ_i is the mean firing rate of a unit the i th bin, λ is the overall mean firing rate, and p_i is the probability of the animal being in the i th bin (occupancy in the i th bin/total occupancies bin).

The single-lap(or trial) rate-maps was calculated using the same method and used to Bayesian decoding analysis.

Local field potential analysis

Theta, slow gamma, and fast gamma rhythms were extracted from the LFP by applying band-pass filters with frequency ranges of 4-12 Hz, 25-45 Hz, and 65-100 Hz, respectively. The time-varying phases of the theta, slow gamma, and fast gamma rhythms were computed using the Hilbert transform. As the previous study showed, the gamma power spectra were measured across different running speeds. The power spectra across running speeds and absolute power spectrum (both result were not shown) was estimated through the multitaper spectral analysis (Mitra and Bokil, 2008) in the Chronux toolbox (<http://chronux.org/>). We defined the gamma episode as the 160ms window with gamma power higher than 3 standard deviations to observe the firing activity of place cells.

Quantification of theta phase precession

Theta phase precessions were examined in the primary place field calculated by all trial spikes. The phase-position correlation of spikes emitted by a place cell during all lap of its place field was determined by circular-linear regression between the theta phase of spikes and the actual position of animal. The significant negative correlation was determined theta phase precession. Theta phase precession slope was calculated by least-square fit of spike locations of the animal and shifted spike phases.

Estimation of gamma phase-locking

Phase-lock analysis was implemented using custom scripts based on the `circ_stats` function from Matlab CircStat toolbox. The slow gamma, and fast gamma spike phase distributions for each place cell were then determined by the time-varying phases of slow gamma, and fast gamma (see “Local field potential analysis”), respectively, at the time point closest to each spike time. And each spike phase was evaluated with the LFP recorded in the tetrode where the neuron was located. Phase-locking was quantified by calculating the mean vector length of the resulting phase distributions. Significant phase-locking at fast gamma is identified only for place cells meeting the criteria of a Rayleigh test $p < 0.05$ and firing at least 10 spikes on each lap. For easy presentation, these cells are named FG-cells. The remaining place cell are not phase-locking at fast gamma, called NFG-cells. According to our results, nearly a quarter of the neurons are significantly phase-locked at fast gamma. Only a very small number of neurons are slow gamma phase-locked.

Single-trial Bayesian decoding analysis

To translate ensemble spiking activity into radian positions on the circular track, a memory-less Bayesian decoding algorithm was implemented. Decoding was performed using a 20ms sliding time window that shifted 5ms at each step. For each decoding window, the number of spikes occurring is denoted as n_i , so that the vector of all spiking activity simultaneously recorded for the cell is $\mathbf{n} = \{n_1, n_2, \dots, n_i\}$. The probability of a rat being at position x , given the number of spikes $\mathbf{n}$ from each unit recorded in a time window, was estimated by Bayes’ rule:

$$P(x|\mathbf{n}) = \frac{P(\mathbf{n}|x) \times P(x)}{P(\mathbf{n})}$$

where $P(\mathbf{n}|x)$ was approximated from the single directional firing turning curve of each unit. The approximation assumed that the turning curve of individual units was statistically independent and the number of spikes from each unit followed a Poisson distribution. Prior knowledge of position $P(x)$ was set to 1 to avoid decoding bias to any particular location on the track. The normalizing constant $P(\mathbf{n})$ was set to ensure $P(x|\mathbf{n})$, or posterior probability, summed to 1. The decoding position was the set of x that maximizes $P(x|\mathbf{n})$. Decoding error was defined as the distance between the decoded position with maximum-likelihood and the rat’s true position. For excluded cell decoding, position reconstruction was performed by removing the FG-cells or NFG-cells from $\mathbf{n}$, respectively. Bayesian decoding analyses were performed using custom routines in MATLAB 2020b.

Theta cycle detection

A single tetrode located in the stratum pyramidale and physical center all recording tetrodes was selected and the theta signal (including phase and power etc.) from this tetrode was utilized for further analysis. According to previous studies, almost all theta sequences fall completely into a theta cycle. Therefore, the global 0 phase (the beginning of the theta cycle) of theta rhythm was determined as the phase when the theta sequence quantization indicators (see “Theta sequence quantification”) were maximal. Briefly, the global 0 phase was moved from the peaks in steps of

10°, and in every step, the sequence of place cells was quantified in each theta cycle. The peak of the histogram of the distribution of the quantization indicators over the step size was used as the global 0 theta phase for further analysis.

Theta sequence quantification

Theta sequences were quantified only when the following conditions also hold: (1) the animal was in the middle of the track (two-thirds of the circular track length), (2) the current running direction is consistent with the expected direction with running faster than 5 cm/s, (3) the duration of the corresponding theta cycles ranged from 100 to 200 ms, (4) more than 3 place cells and emitted at least 5 spikes during the corresponding theta cycle.

Averaged sequence was constructed over the laps by aligning the probabilities over the theta cycle mid-time point and position. Each averaged sequence were quantified within an area that decoded probabilities ± 35 cm around the position of animal, and $\pm 1/4$ theta cycle around the mid-time point of the theta sequence.

Probability difference

The area used to quantify were divided equally into four quadrants. The Quadrant 2 represents the region both physically and temporally ahead of the rat, whereas as the Quadrant 3 represents the region physically and temporally behind the rat. Quadrant 1 and 4 represents the region that opposite with the animal's current running direction. The summed decoded probabilities in Quadrant 1 and 4 were subtracted from the summed decoded probabilities in Quadrant 2 and 3 then divided by the summed decoded probabilities in all four quadrants. This value represents the difference between the decoding probability of the normalized current path and the opposite path in the theta sequence. Positive differences would imply theta sequences sweeping ahead in current moving direction of the rat, whereas differences close to zero would indicate a lack of sequential structure.

Weighted correlation

Weighted correlation of each or averaged theta sequence was calculated from the same area mentioned above. With the decoding probability (*prob*) as the weight of the position estimation, the correlation coefficient between the time (*t*) and the decoding position (*p*) is as follows:

$$corr_w(t, p; prob) = \frac{cov(t, p; prob)}{\sqrt{cov(t, t; prob)cov(p, p; prob)}}$$

where weighted covariance between time and decoded position is as follows:

$$cov(t, p; prob) = \frac{\sum_i prob_i (t_i - m(t; prob))(p_i - m(p; prob))}{\sum_i prob_i}$$

and weighted means of time and decoded position are as follows:

$$m(t; prob) = \frac{\sum_i prob_i t_i}{\sum_i prob_i}$$

and

$$m(p; prob) = \frac{\sum_i prob_i p_i}{\sum_i prob_i}$$

The apparent sweep-ahead structure in the current running direction of animal produced large positive correlations (close to one), whereas the lack of sequence structure produced close to zero correlations. For further analysis, we use weighted correlation to quantify the theta sequence from different decoding result.

Flip correlation difference

We defined an Indicator to measure the front-sweep structure, named Flip correlation difference, which of each or averaged theta sequence was calculated also from the same area mentioned above. In the temporal dimension, the correlation coefficient between the decoding probability and the decoding probability after performing both temporal and spatial flips was calculated, from which the correlation coefficient between the decoding probability and the decoding probability after performing only temporal flips was subtracted. Positive difference represents intact sequential structure in the positive, whereas differences close to zero or negative would indicate a lack of sequential structure or defective sequential structure.

Gamma phase shift analysis of cell ensemble

The gamma cycle with maximal spiking across all simultaneously recorded cells was defined as cycle 0. Gamma cycles occurring before cycle 0 are represented by decreasing integer values, after by increasing integer values. Incomplete cycles at the beginning or end of the sequence or during immobile (speed < 5 cm/s) were excluded from analyses. We estimated the gamma phases and theta phases of spike times for place cells that exhibited spikes within at least two slow gamma cycles or at least three fast gamma cycles within a single theta cycle. The number of cycles analyzed within each theta sequence was limited to three (cycles -1 to 1) for slow gamma and five (cycles -2 to 2) for fast gamma (data not shown). Two-dimensional histograms of gamma phases and theta phases for spikes from each cycle number were binned into 40 bins and smoothed across nine bins.

Slow gamma phase shift analysis of individual cell

We employed Morlet's wavelet method to calculate the time-varying power of slow gamma. The slow gamma energy in each theta cycle was estimated, considering only cycles with energy higher than 1.5 times the standard deviation as slow gamma events. The global 0 phase of slow gamma, set at the trough, determined the phase shift. For each cell, spikes in different slow gamma cycles within each theta cycle were identified. Spike pairs, representing adjacent spikes from different slow gamma cycles, were used to calculate the average phase difference as the slow gamma phase difference. In cases of multiple spikes in the same slow gamma cycle, their average phase was taken. Negative values indicated phase precession, while positive values indicated procession. To maintain firing location constancy, spikes used for phase difference calculation were assigned a random slow gamma phase within their cycle, repeated 1000 times for a distribution. In the analysis, we calculated one-sided, corrected empirical p-values $[(r+1)/(n+1)]$ by comparing real data to its corresponding shuffle data. Where r is the number values from the shuffled distribution that are either smaller or larger (depending on the hypothesis) than the observed value and $n=1000$ is the number of shuffling.

Correlation between the slow gamma phase precession and weighted correlation

The theta cycle satisfying the criteria (see Slow gamma phase shift analysis) for calculating the phase difference was chosen to estimate the relationship between slow gamma phase precession and weighted correlation. On the one hand, the Pearson correlation coefficient was used to estimate the linear correlation between them. On the other hand, the correlation between slow gamma phase precession and theta sequence structure was analyzed by two-dimensional histogram. We binned slow gamma phase difference and weighted correlation into ten bins to construct a two-dimensional histogram, and smoothed it across three bins for visualize. The Two-dimensional histogram was divided into four quadrants and the sum of probability in Quadrant 1 (Q1) was calculated. The statistical significance was determined by calculating the one-sided, corrected empirical p-value. The shuffle data was same as slow gamma phase shuffle. The true

data was considered to be significantly concentrated in Q1. The null hypothesis was the probability in Q1 of shuffled data was larger than the true data. If the p value was less than 0.05, the null hypothesis was rejected.

Histology

At the end of the experiment, histological sections were prepared as follows to verify the tetrodes position. Rats were intraperitoneally injected with a lethal dose of urethane solution. The heart was subsequently perfused with phosphate buffer and formalin to fix the brain tissue. Rat brains were then cut into 30- μ m coronal sections and Nissl Staining was performed to determine the location of tetrode in CA1.

Quantification and statistical analysis

Data analyses were performed using SPSS (IBM) and custom Matlab (the Math Work) and R scripts. Paired t-test was performed to test between exFG-sequences and exNFG-sequences during early and late development (**Fig.3H** [↗](#)). Student's t test was performed to test the spike number, mean firing rate in field, place fields size and spatial information between the FG-cell and NFG-cell (**Fig.5D-G** [↗](#)). One-sample t test was conducted to assess whether the slow gamma phase shift was significantly lower than 0 (**Fig.7B** [↗](#)). Repeated measures ANOVA was performed to test the weighted correlation of theta sequences (**Fig.2D** [↗](#), **Fig.3C,E,G** [↗](#)) and running speed (**Fig.3F** [↗](#)) as a function of laps. Kolmogorov-Smirnov test was performed to test the distribution of place field COM (**Fig.5C** [↗](#)) on the track using the Matlab function 'kstest2'. Multi-sample Mardia-Watson-Wheeler test was performed to test the slow gamma phase shift of place cells across successive slow gamma cycles (**Fig.6** [↗](#)) using the R 'circular-package'. Pearson's correlation was performed to test the correlation between weighted correlation difference and the proportion of excluded cells (**Fig.4C-D** [↗](#)).

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Author contributions

Conceptualization, C.Z. and L.W.; Methodology and Software, N.W., L.W. and C.Z.; Investigation, N.W., Y.W., M.G., L.W. N.Z., X.W. and C.Z.; Formal Analysis, N.W. and L.W.; Writing – Original Draft, N.W. and C.Z.; Funding Acquisition, C.Z., M.D., L.W., J.Y., and L.W.; Supervision, C.Z. and D.M.

Competing interests

The authors declare that they have no competing interests.

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Editors

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Reviewer #1 (Public Review):

Hippocampal place cells display a sequence of firing activities when the animal travels through a spatial trajectory at a behavioral time scale of seconds to tens of seconds. Interestingly, parts of the firing sequence also occur at a much shorter time scale: ~120 ms within individual cycles of theta oscillation. These so-called theta sequences are originally thought to naturally result from the phenomenon of theta phase precession. However, there is evidence that theta sequences do not always occur even when theta phase precession is present, for example, during the early experience of a novel maze. The question is then how they emerge with experience (theta sequence development). This study presents evidence that a special group of place cells, those tuned to fast-gamma oscillations, may play a key role in theta sequence development.

The authors analyzed place cells, LFPs, and theta sequences as rats traveled a circular maze in repeated laps. They found that a group of place cells were significantly tuned to a particular phase of fast-gamma (FG-cells), in contrast to others that did not show such tuning (NFG-cells). The authors then omitted FG-cells or the same number of NFG-cells, in their algorithm of theta sequence detection and found that the quality of theta sequences, quantified by a weighted correlation, was worse with the FG-cell omission, compared to that with the NFG-cell omission, during later laps, but not during early laps. What made the FG-cells special for theta sequences? The authors found that FG-cells, but not NFG-cells, displayed phase recession to slow-gamma (25 - 45 Hz) oscillations (within theta cycles) during early laps (both FG- and NFG-cells showed slow-gamma phase precession during later laps). Overall, the authors conclude that FG-cells contribute to theta sequence development through slow-gamma phase precession during early laps.

How theta sequences are formed and developed during experience is an important question, because these sequences have been implicated in several cognitive functions of place cells, including memory-guided spatial navigation. The identification of FG-cells in this study is straightforward. Evidence is also presented for the role of these cells in theta sequence development. However, given several concerns elaborated below, whether the evidence is sufficiently strong for the conclusion needs further clarification, perhaps, in future studies.

(1) The results in Figure 3 and Figure 8 seems contradictory. In Figure 8, all theta sequences displayed a seemingly significant weighted correlation (above 0) even in early laps, which was mostly due to FG-cell sequences but not NFG-cell sequences (correlation for NFG-sequences appeared below 0). However, in Figure 3H, omitting FG-cells and omitting NFG-cells did not produce significant differences in the correlation. Conversely, FG-cell and NFG-cell sequences were similar in later laps in Figure 8 (NFG-cell sequences appeared even better than FG-cell sequences), yet omitting NFG-cells produced a better correlation than omitting FG-cells. This confusion may be related to how "FG-cell-dominant sequences" were defined, which is unclear in the manuscript. Nevertheless, the different results are not easy to understand.

(2) The different contributions between FG-cells and NFG-cells to theta sequences are supposed not to be caused by their different firing properties (Figure 5). However, Figure 5D and E showed a large effect size (Cohen's $D = 0.7, 0.8$), although not significant ($P = 0.09, 0.06$). But the seemingly non-significant P values could be simply due to smaller N 's (~ 20). In other parts of the manuscript, the effect sizes were comparable or even smaller (e.g. $D = 0.5$ in Figure 7B), but interpreted as positive results: P values were significant with large N 's (~ 480 in Fig. 7B). Drawing a conclusion purely based on a P value while N is large often renders the conclusion only statistical, with unclear physical meaning. Although this is common in neuroscience publications, it makes more sense to at least make multiple inferences using similar sample sizes in the same study.

(3) In supplementary Figure 2 - S2, FG-cells displayed stronger theta phase precession than NFG-cells, which could be a major reason why FG-cells impacted theta sequences more than NFG cells. Although factors other than theta phase precession may contribute to or interfere with theta sequences, stronger theta phase precession itself (without the interference of other factors), by definition, can lead to stronger theta sequences.

(4) The slow-gamma phase precession of FG-cells during early laps is supposed to mediate or contribute to the emergence of theta sequences during late laps (Figure 1). The logic of this model is unclear. The slow-gamma phase precession was present in both early and late laps for FG-cells, but only present in late laps for NFG-cells. It seems more straightforward to hypothesize that the difference in theta sequences between early and later laps is due to the difference in slow-gamma phase precession of NFG cells between early and late laps. Although this is not necessarily the case, the argument presented in the manuscript is not easy to follow.

(5) There are several questions on the description of methods, which could be addressed to clarify or strengthen the conclusions.

(i) Were the identified fast- and slow-gamma episodes mutually exclusive?

(ii) Was the task novel when the data were acquired? How many days (from the 1st day of the task) were included in the analysis? When the development of the theta sequence was mentioned, did it mean the development in a novel environment, in a novel task, or purely in a sense of early laps (Lap 1, 2) on each day?

(iii) How were the animals' behavioral parameters equalized between early and later laps? For example, speed or head direction could potentially produce the differences in theta sequences.

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Reviewer #2 (Public Review):

This manuscript addresses an important question that has not yet been solved in the field, what is the contribution of different gamma oscillatory inputs to the development of "theta sequences" in the hippocampal CA1 region? Theta sequences have received much attention due to their proposed roles in encoding short-term behavioral predictions, mediating synaptic plasticity, and guiding flexible decision-making. Gamma oscillations in CA1 offer a readout of different inputs to this region and have been proposed to synchronize neuronal assemblies and modulate spike timing and temporal coding. However, the interactions between these two important phenomena have not been sufficiently investigated. The authors conducted place cell and local field potential (LFP) recordings in the CA1 region of rats running on a circular track. They then analyzed the phase locking of place cell spikes to slow and fast gamma rhythms, the evolution of theta sequences during behavior, and the

interaction between these two phenomena. They found that place cells with the strongest modulation by fast gamma oscillations were the most important contributors to the early development of theta sequences and that they also displayed a faster form of phase precession within slow gamma cycles nested with theta. The results reported are interesting and support the main conclusions of the authors. However, the manuscript needs significant improvement in several aspects regarding data analysis, description of both experimental and analytical methods, and alternative interpretations, as I detail below.

- The experimental paradigm and recordings should be explained at the beginning of the Results section. Right now, there is no description whatsoever which makes it harder to understand the design of the study.
- An important issue that needs to be addressed is the very small fraction of CA1 cells phased-locked to slow gamma rhythms (3.7%). This fraction is much lower than in many previous studies, that typically report it in the range of 20-50 %. However, this discrepancy is not discussed by the authors. This needs to be explained and additional analysis considered. One analysis that I would suggest, although there are also other valid approaches, is to, instead of just analyzing the phase locking in two discrete frequency bands, compute the phase locking will all LFP frequencies from 25-100 Hz. This will offer a more comprehensive and unbiased view of the gamma modulation of place cell firing. Alternative metrics to mean vector length that is less sensitive to firing rates, such as pairwise phase consistency index (Vinck et al., Neuroimage, 2010), could be implemented. This may reveal whether the low fraction of phase-locked cells could be due to a low number of spikes entering the analysis.
- From the methods, it is not clear to me whether the reference LFP channel was consistently selected to be a different one that where the spikes analyzed were taken. This is the better practice to reduce the contribution of spike leakage that could substantially inflate the coupling with faster gamma frequencies. These analyses need to be described in more detail.
- The initial framework of the authors of classifying cells into fast gamma and not fast gamma modulated implies a bimodality that may be artificial. The authors should discuss the nuances and limitations of this framework. For example, several previous work has shown that the same place cell can couple to different gamma oscillations (e.g., Lastoczni et al., Neuron, 2016; Fernandez-Ruiz et al., Neuron, 2017; Sharif et al., Neuron, 2021).
- It would be useful to provide a more thorough characterization of the physiological properties of FG and NFG cells, as this distinction is the basis of the paper. Only very little characterization of some place cell properties is provided in Figure 5. Important characteristics that should be very feasible to compare include average firing rate, burstiness, estimated location within the layer (i.e., deep vs superficial sublayers) and along the transverse axis (i.e., proximal vs distal), theta oscillation frequency, phase precession metrics (given their fundamental relationship with theta sequences), etc.
- It is not clear to me how the analysis in Figure 6 was performed. In Figure 6B I would think that the grey line should connect with the bottom white dot in the third panel, which would be the interpretation of the results.

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Reviewer #3 (Public Review):

[Editors' note: This review contains many criticisms that apply to the whole sub-field of slow/fast gamma oscillations in the hippocampus, as opposed to this particular paper. In the editors' view, these comments are beyond the scope of any single paper. However, they represent a view that, if true, should contextualise the interpretation of this paper and all

papers in the sub-field. In doing so, they highlight an ongoing debate within the broader field.]

Summary:

The authors aimed to elucidate the role of dynamic gamma modulation in the development of hippocampal theta sequences, utilizing the traditional framework of "two gammas," a slow and a fast rhythm. This framework is currently being challenged, necessitating further analyses to establish and secure the assumed premises before substantiating the claims made in the present article.

The results are too preliminary and need to integrate contemporary literature. New analyses are required to address these concerns. However, by addressing these issues, it may be possible to produce an impactful manuscript.

I. Introduction

Within the introduction, multiple broad assertions are conveyed that serve as the premise for the research. However, equally important citations that are not mentioned potentially contradict the ideas that serve as the foundation. Instances of these are described below:

(1) Are there multiple gammas? The authors launched the study on the premise that two different gamma bands are communicated from CA3 and the entorhinal cortex. However, recent literature suggests otherwise, offering that the slow gamma component may be related to theta harmonics:

From a review by Etter, Carmichael and Williams (2023)

"Gamma-based coherence has been a prominent model for communication across the hippocampal-entorhinal circuit and has classically focused on slow and fast gamma oscillations originating in CA3 and medial entorhinal cortex, respectively. These two distinct gammas are then hypothesized to be integrated into hippocampal CA1 with theta oscillations on a cycle-to-cycle basis (Colgin et al., 2009; Schomburg et al., 2014). This would suggest that theta oscillations in CA1 could serve to partition temporal windows that enable the integration of inputs from these upstream regions using alternating gamma waves (Vinck et al., 2023). However, these models have largely been based on correlations between shifting CA3 and medial entorhinal cortex to CA1 coherence in theta and gamma bands. In vivo, excitatory inputs from the entorhinal cortex to the dentate gyrus are most coherent in the theta band, while gamma oscillations would be generated locally from presumed local inhibitory inputs (Pernía-Andrade and Jonas, 2014). This predominance of theta over gamma coherence has also been reported between hippocampal CA1 and the medial entorhinal cortex (Zhou et al., 2022). Another potential pitfall in the communication-through-coherence hypothesis is that theta oscillations harmonics could overlap with higher frequency bands (Czurkó et al., 1999; Terrazas et al., 2005), including slow gamma (Petersen and Buzsáki, 2020). The asymmetry of theta oscillations (Belluscio et al., 2012) can lead to harmonics that extend into the slow gamma range (Scheffer-Teixeira and Tort, 2016), which may lead to a misattribution as to the origin of slow-gamma coherence and the degree of spike modulation in the gamma range during movement (Zhou et al., 2019)."

And from Benjamin Griffiths and Ole Jensen (2023)

"That said, in both rodent and human studies, measurements of 'slow' gamma oscillations may be susceptible to distortion by theta harmonics [53], meaning open questions remain about what can be attributed to 'slow' gamma oscillations and what is attributable to theta."

This second statement should be heavily considered as it is from one of the original authors who reported the existence of slow gamma.

Yet another instance from Schomburg, Fernández-Ruiz, Mizuseki, Berényi, Anastassiou, Christof Koch, and Buzsáki (2014):

"Note that modulation from 20-30 Hz may not be related to gamma activity but, instead, reflect timing relationships with non-sinusoidal features of theta waves (Belluscio et al., 2012) and/or the 3rd theta harmonic."

One of this manuscript's authors is Fernández-Ruiz, a contemporary proponent of the multiple gamma theory. Thus, the modulation to slow gamma offered in the present manuscript may actually be related to theta harmonics.

With the above emphasis from proponents of the slow/fast gamma theory on disambiguating harmonics from slow gamma, our first suggestion to the authors is that they A) address these statements (citing the work of these authors in their manuscript) and B) demonstrably quantify theta harmonics in relation to slow gamma prior to making assertions of phase relationships (methodological suggestions below). As the frequency of theta harmonics can extend as high as 56 Hz (PMID: 32297752), overlapping with the slow gamma range defined here (25-45 Hz), it will be important to establish an approach that decouples the two phenomena using an approach other than an arbitrary frequency boundary.

(2) Can gammas be segregated into different lamina of the hippocampus? This idea appears to be foundational in the premise of the research but is also undergoing revision.

As discussed by Etter et al. above, the initial theory of gamma routing was launched on coherence values. However, the values reported by Colgin et al. (2009) lean more towards incoherence (a value of 0) rather than coherence (1), suggesting a weak to negligible interaction. Nevertheless, this theory is coupled with the idea that the different gamma frequencies are exclusive to the specific lamina of the hippocampus.

Recently, Deschamps et al. (2024) suggested a broader, more nuanced understanding of gamma oscillations than previously thought, emphasizing their wide range and variability across hippocampal layers. This perspective challenges the traditional dichotomy of gamma sub-bands (e.g., slow vs. medium gamma) and their associated cognitive functions based on a more rigid classification according to frequency and phase relative to the theta rhythm. Moreover, they observed all frequencies across all layers.

Similarly, the current source density plots from Belluscio et al. (2012) suggest that SG and FG can be observed in both the radiatum and lacunosum-moleculare.

Therefore, if the initial coherence values are weak to negligible and both slow and fast gamma are observed in all layers of the hippocampus, can the different gammas be exclusively related to either anatomical inputs or psychological functions (as done in the present manuscript)? Do these observations challenge the authors' premise of their research? At the least, please discuss.

(3) Do place cells, phase precession, and theta sequences require input from afferent regions? It is offered in the introduction that "Fast gamma (~65-100Hz), associated with the input from the medial entorhinal cortex, is thought to rapidly encode ongoing novel information in the context (Fernandez-Ruiz et al., 2021; Kemere, Carr, Karlsson, & Frank, 2013; Zheng et al., 2016)".

CA1 place fields remain fairly intact following MEC inactivation include Ipshita Zutshi, Manuel Valero, Antonio Fernández-Ruiz, and György Buzsáki (2022)- "CA1 place cells and assemblies persist despite combined mEC and CA3 silencing" and from Hadas E Sloin, Lidor Spivak, Amir Levi, Roni Gattegno, Shirley Someck, Eran Stark (2024) - "These findings are incompatible with precession models based on inheritance, dual-input, spreading activation, inhibition-excitation summation, or somato-dendritic competition. Thus, a precession generator resides locally within CA1."

These publications, at the least, challenge the inheritance model by which the afferent input controls CA1 place field spike timing. The research premise offered by the authors is couched in the logic of inheritance, when the effect that the authors are observing could be governed by local intrinsic activity (e.g., phase precession and gamma are locally generated, and the attribution to routed input is perhaps erroneous). Certainly, it is worth discussing these manuscripts in the context of the present manuscript.

II. Results

(1) Figure 2-

a. There is a bit of a puzzle here that should be discussed. If slow and fast frequencies modulate 25% of neurons, how can these rhythms serve as mechanisms of communication/support psychological functions? For instance, if fast gamma is engaged in rapid encoding (line 72) and slow gamma is related to the integration processing of learned information (line 84), and these are functions of the hippocampus, then why do these rhythms modulate so few cells? Is this to say 75% of CA1 neurons do not listen to CA3 or MEC input?

b. Figure 2. It is hard to know if the mean vector lengths presented are large or small. Moreover, one can expect to find significance due to chance. For instance, it is challenging to find a frequency in which modulation strength is zero (please see Figure 4 of PMID: 30428340 or Figure 7 of PMID: 31324673).

i. Please construct the histograms of Mean Vector Length as in the above papers, using 1 Hz filter steps from 1-120Hz and include it as part of Figure 2 (i.e., calculate the mean vector length for the filtered LFP in steps of 1-2 Hz, 2-3 Hz, 3-4 Hz,... etc). This should help the authors portray the amount of modulation these neurons have relative to the theta rhythm and other frequencies. If the theta mean vector length is higher, should it be considered the primary modulatory influence of these neurons (with slow and fast gammas as a minor influence)?

ii. It is possible to infer a neuron's degree of oscillatory modulation without using the LFP. For instance, one can create an ISI histogram as done in Figure 1 here (<https://www.biorxiv.org/content/10.1101/2021.09.20.461152v3.full.pdf+html>; "Distinct ground state and activated state modes of firing in forebrain neurons"). The reciprocal of the ISI values would be "instantaneous spike frequency". In favor of the Douchamps et al. (2024) results, the figure of the BioRxiv paper implies that there is a single gamma frequency modulate as there is only a single bump in the ISIs in the $10^{-1.5}$ to 10^{-2} range. Therefore, to vet the slow gamma results and the premise of two gammas offered in the introduction, it would be worth including this analysis as part of Figure 2.

c. There are some things generally concerning about Figure 2.

i. First, the raw trace does not seem to have clear theta epochs (it is challenging to ascertain the start and end of a theta cycle). Certainly, it would be worth highlighting the relationship between theta and the gammas and picking a nice theta epoch.

ii. Also, in panel A, there looks to be a declining amplitude relationship between the raw, fast, and slow gamma traces, assuming that the scale bars represent 100uV in all three traces. The raw trace is significantly larger than the fast gamma. However, this relationship does not seem to be the case in panel B (in which both the raw and unfiltered examples of slow and fast gamma appear to be equal; the right panels of B suggest that fast gamma is larger than slow, appearing to contradict the $A = 1/f$ organization of the power spectral density). Please explain as to why this occurs. Including the power spectral density (see below) should resolve some of this.

iii. Within the example of spiking to phase in the left side of Panel B (fast gamma example)- the neuron appears to fire near the trough twice, near the peak twice, and somewhere in between once. A similar relationship is observed for the slow gamma epoch. One would conclude from these plots that the interaction of the neuron with the two rhythms is the same. However, the mean vector lengths and histograms below these plots suggest a different story in which the neuron is modulated by FG but not SG. Please reconcile this.

iv. For calculating the MVL, it seems that the number of spikes that the neuron fires would play a significant role. Working towards our next point, there may be a bias of finding a relationship if there are too few spikes (spurious clustering due to sparse data) and/or higher coupling values for higher firing rate cells (cells with higher firing rates will clearly show a relationship), forming a sort of inverse Yerkes-Dodson curve. Also, without understanding the magnitude of the MVL relative to other frequencies, it may be that these values are indeed larger than zero, but not biologically significant.

- Please provide a scatter plot of Neuron MVL versus the Neuron's Firing Rate for 1) theta (7-9 Hz), 2) slow gamma, and 3) fast gamma, along with their line of best fit.

- Please run a shuffle control where the LFP trace is shifted by random values between 125-1000ms and recalculate the MVL for theta, slow, and fast gamma. Often, these shuffle controls are done between 100-1000 times (see cross-correlation analyses of Fujisawa, Buzsaki et al.).

- To establish that firing rate does not play a role in uncovering modulation, it would be worth conducting a spike number control, reducing the number of spikes per cell so that they are all equal before calculating the phase plots/MVL.

(2) Something that I anticipated to see addressed in the manuscript was the study from Grosmark and Buzsaki (2016): "Cell assembly sequences during learning are "replayed" during hippocampal ripples and contribute to the consolidation of episodic memories. However, neuronal sequences may also reflect preexisting dynamics. We report that sequences of place-cell firing in a novel environment are formed from a combination of the contributions of a rigid, predominantly fast-firing subset of pyramidal neurons with low spatial specificity and limited change across sleep-experience-sleep and a slow-firing plastic subset. Slow-firing cells, rather than fast-firing cells, gained high place specificity during exploration, elevated their association with ripples, and showed increased bursting and temporal coactivation during postexperience sleep. Thus, slow- and fast-firing neurons, although forming a continuous distribution, have different coding and plastic properties."

My concern is that much of the reported results in the present manuscript appear to recapitulate the observations of Grosmark and Buzsaki, but without accounting for differences in firing rate. A parsimonious alternative explanation for what is observed in the present manuscript is that high firing rate neurons, more integrated into the local network and orchestrating local gamma activity (PING), exhibit more coupling to theta and gamma. In this alternative perspective, it's not something special about how the neurons are entrained to the routed fast gamma, but that the higher firing rate neurons are better able to engage and entrain their local interneurons and, thus modulate local gamma. However, this interpretation challenges the discussion around the importance of fast gamma routed from the MEC.

a. Please integrate the Grosmark & Buzsaki paper into the discussion.

b. Also, please provide data that refutes or supports the alternative hypothesis in which the high firing rate cells are just more gamma modulated as they orchestrate local gamma activity through monosynaptic connections with local interneurons (e.g., Marshall et al., 2002, Hippocampal pyramidal cell-interneuron spike transmission is frequency dependent and responsible for place modulation of interneuron discharge). Otherwise, the attribution to a

MEC routed fast gamma routing seems tenuous.

c. It is mentioned that fast-spiking interneurons were removed from the analysis. It would be worth including these cells, calculating the MVL in 1 Hz increments as well as the reciprocal of their ISIs (described above).

(3) Methods - Spectral decomposition and Theta Harmonics.

a. It is challenging to interpret the exact parameters that the authors used for their multi-taper analysis in the methods (lines 516-526). Tallon-Baudry et al., (1997; Oscillatory γ -Band (30-70 Hz) Activity Induced by a Visual Search Task in Humans) discuss a time-frequency trade-off where frequency resolution changes with different temporal windows of analysis. This trade-off between time and frequency resolution is well known as the uncertainty principle of signal analysis, transcending all decomposition methods. It is not only a function of wavelet or FFT, and multi-tapers do not directly address this. (The multitaper method, by using multiple specially designed tapers -like the Slepian sequences- smooths the spectrum. This smoothing doesn't eliminate leakage but distributes its impact across multiple estimates). Given the brevity of methods and the issues of theta harmonics as offered above, it is worth including some benchmark trace testing for the multi-taper as part of the supplemental figures.

i. Please spectrally decompose an asymmetric 8 Hz sawtooth wave showing the trace and the related power spectral density using the multiple taper method discussed in the methods.

ii. Please also do the same for an elliptical oscillation (perfectly symmetrical waves, but also capable of casting harmonics). Matlab code on how to generate this time series is provided below:

```
A = 1; % Amplitude
T = 1/8; % Period corresponding to 8 Hz frequency
omega = 2*pi/T; % Angular frequency
C = 1; % Wave speed
m = 0.9; % Modulus for the elliptic function (0
x = linspace(0, 2*pi, 1000); % temporal domain
t = 0; % Time instant

% Calculate B based on frequency and speed
B = sqrt(omega/C);

% Cnoidal wave equation using the Jacobi elliptic function
u = A .* ellipj(B.*(x - C*t), m).^2;

% Plotting the cnoidal wave
figure;
plot(x./max(x), u);
title('8 Hz Cnoidal Wave');
xlabel('time (x)');
ylabel('Wave amplitude (u)');
grid on;
```

The Symbolic Math Toolbox needs to be installed and accessible in your MATLAB environment to use `ellipj`. Otherwise, I trust that, rather than plotting a periodic orbit around a circle (sin wave) the authors can trace the movement around an ellipse with significant eccentricity (the distance between the two foci should be twice the distance between the co-vertices).

iii. Line 522: "The power spectra across running speeds and absolute power spectrum (both results were not shown)..." Given the potential complications of multi-taper discussed above, and as each convolution further removes one from the raw data, it would be the most

transparent, simple, and straightforward to provide power spectra using the simple `fft.m` code in Matlab (We imagine that the authors will agree that the results should be robust against different spectral decomposition methods. Otherwise, it is concerning that the results depend on the algorithm implemented and should be discussed. If gamma transience is a concern, the authors should trigger to 2-second epochs in which slow/fast gamma exceeds 3-7 std. dev. above the mean, comparing those resulting power spectra to 2-second epochs with ripples - also a transient event). The time series should be at least 2 seconds in length (to avoid spectral leakage issues and the issues discussed in Talon-Baudry et al., 1997 above).

Please show the unmolested power spectra (Y-axis units in mV^2/Hz , X-axis units as Hz) as a function of running speed (increments of 5 cm/s) for each animal. I imagine three of these PSDs for 3 of the animals will appear in supplemental methods while one will serve as a nice manuscript figure. With this plot, please highlight the regions that the authors are describing as theta, slow, and fast gamma. Also, any issues should be addressed should there be notable differences in power across animals or tetrodes (issues with locations along proximal-distal CA1 in terms of MEC/LEC input and using a local reference electrode are discussed below).

iv. Schomberg and colleagues (2014) suggested that the modulation of neurons in the slow gamma range could be related to theta harmonics (see above). Harmonics can often extend in a near infinite as they regress into the $1/f$ background (contributing to power, but without a peak above the power spectral density slope), making arbitrary frequency limits inappropriate. Therefore, in order to support the analyses and assertions regarding slow gamma, it seems necessary to calculate a "theta harmonic/slow gamma ratio". Aru et al. (2015; Untangling cross-frequency coupling in neuroscience) offer that: "The presence of harmonics in the signal should be tested by a bicoherence analysis and its contribution to CFC should be discussed." Please test both the synthetic signals above and the raw LFP, using temporal windows of greater than 4 seconds (again, the large window optimizes for frequency resolution in the time-frequency trade-off) to calculate the bicoherence. As harmonics are integers of theta coupled to itself and slow gamma is also coupled to theta, a nice illustration and contribution to the field would be a method that uses the bispectrum to isolate and create a "slow gamma/harmonic" ratio.

(4) I appreciate the inclusion of the histology for the 4 animals. Knerim and colleagues describe a difference in MEC projection along the proximal-distal axis of the CA1 region (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3866456/>) - "There are also differences in their direct projections along the transverse axis of CA1, as the LEC innervates the region of CA1 closer to the subiculum (distal CA1), whereas the MEC innervates the region of CA1 closer to CA2 and CA3 (proximal CA1)" From the histology, it looks like some of the electrodes are in the part of CA1 that would be dominated by LEC input while a few are closer to where the MEC would project.

a. How do the authors control for these differences in projections? Wouldn't this change whether or not fast gamma is observed in CA1?

b. I am only aware of one manuscript that describes slow gamma in the LEC which appeared in contrast to fast gamma from the MEC (<https://www.science.org/doi/10.1126/science.abf3119>). One would surmise that the authors in the present manuscript would have varying levels of fast gamma in their CA1 recordings depending on the location of the electrodes in the Proximal-distal axis, to the extent that some of the more medial tetrodes may need to be excluded (as they should not have fast gamma, rather they should be exclusively dominated by slow gamma). Alternatively, the authors may find that there is equal fast gamma power across the entire proximal-distal axis. However, this would pose a significant challenge to the LEC/slow gamma and MEC/fast gamma routing story of Fernandez-Ruiz et al. and require reconciliation/discussion.

c. Is there a difference in neuron modulation to these frequencies based on electrode location in CA1?

(5) Given a comment in the discussion (see below), it will be worth exploring changes in theta, theta harmonic, slow gamma, and fast gamma power with running speed as no changes were observed with theta sequences or lap number versus. Notably, Czurko et al., report an increase in theta and harmonic power with running speed (1999) while Ahmed and Mehta (2012) report a similar effect for gamma.

a. Please determine if the oscillations change in power and frequency of the rhythms discussed above change with running speed using the same parameters applied in the present manuscript. The specific concern is that how the authors calculate running speed is not sensitive enough to evaluate changes.

b. It is astounding that animals ran as fast as they did in what appears to be the first lap (Figure 3F), especially as rats' natural proclivity is thigmotaxis and inquisitive exploration in novel environments. Can the authors expand on why they believe their rats ran so quickly on the first lap in a novel environment and how to replicate this? Also, please include the individual values for each animal on the same plot.

c. Can the authors explain how the statistics on line 169 ($F(4,44)$) work? Specifically, it is challenging to determine how the degrees of freedom were calculated in this case and throughout if there were only 4 animals (reported in methods) over 5 laps (depicted in Figure 3F. Given line 439, it looks like trials and laps are used synonymously). Four animals over 5 laps should have a DOF of 16.

(6) Throughout the manuscript, I am concerned about an inflation of statistical power. For example on line 162, $F(2,4844)$. The large degrees of freedom indicate that the sample size was theta sequences or a number of cells. Since multiple observations were obtained from the same animal, the statistical assumption of independence is violated. Therefore, the stats need to be conducted using a nested model as described in Aarts et al. (2014; <https://pubmed.ncbi.nlm.nih.gov/24671065/>). A statistical consult may be warranted.

(7) It is stated that one tetrode served as a quiet recording reference. The "quiet" part is an assumption when often, theta and gamma can be volume conducted to the cortex (e.g., Sirota et al., 2008; This is often why laboratories that study hippocampal rhythms use the cerebellum for the differential recording electrode and not an electrode in the corpus callosum). Generally, high frequencies propagate as well as low frequencies in the extracellular milieu (<https://www.eneuro.org/content/4/1/ENEURO.0291-16.2016>). For transparency, the authors should include a limitation paragraph in their discussion that describes how their local tetrode reference may be inadvertently diminishing and/or distorting the signal that they are trying to isolate. Otherwise, it would be worth hearing an explanation as to how the author's approach avoids this issue.

Apologetically, this review is already getting long. Moreover, I have substantial concerns that should be resolved prior to delving into the remainder of the analyses. e.g., the analyses related to Figure 3-5 assert that FG cells are important for sequences. However, the relationship to gamma may be secondary to either their relationship to theta or, based on the Grosmark and Buzsaki paper, it may just be a phenomenon coupled to the fast-firing cells (fast-firing cells showing higher gamma modulation due to a local PING dynamic). Moreover, the observation of slow gamma is being challenged as theta harmonics, even by the major proponents of the slow/fast gamma theory. Therefore, the report of slow gamma precession would come as an unsurprising extension should they be revealed to be theta harmonics (however, no control for harmonics was implemented; suggestions were made above). Following these amendments, I would be grateful for the opportunity to provide further feedback.

III. Discussion.

a. Line 330- it was offered that fast gamma encodes information while slow gamma integrates in the introduction. However, in a task such as circular track running (from the methods, it appears that there is no new information to be acquired within a trial), one would guess that after the first few laps, slow gamma would be the dominant rhythm. Therefore, one must wonder why there are so few neurons modulated by slow gamma (~3.7%).

b. Line 375: The authors contend that: "...slow gamma, related to information compression, was also required to modulate fast gamma phase-locked cells during sequence development. We replicated the results of slow gamma phase precession at the ensemble level (Zheng et al., 2016), and furthermore observed it at late development, but not early development, of theta sequences." In relation to the idea that slow gamma may be coupled to - if not a distorted representation of - theta harmonics, it has been observed that there are changes in theta relative to novelty.

i. A. Jeewajee, C. Lever, S. Burton, J. O'Keefe, and N. Burgess (2008) report a decrease in theta frequency in novel circumstances that disappears with increasing familiarity.

ii. One could surmise that this change in frequency is associated with alterations in theta harmonics (observed here as slow gamma), challenging the author's interpretation.

iii. Therefore, the authors have a compelling opportunity to replicate the results of Jeewajee et al., characterizing changes of theta along with the development of slow gamma precession, as the environment becomes familiar. It will become important to demonstrate, using bicoherence as offered by Aru et al., how slow gamma can be disambiguated from theta harmonics. Specifically, we anticipate that the authors will be able to quantify A) theta harmonics (the number, and their respective frequencies and amplitudes), B) the frequency and amplitude of slow gamma, and C) how they can be quantitatively decoupled. Through this, their discussion of oscillatory changes with novelty-familiarity will garner a significant impact.

c. Broadly, it is interesting that the authors emphasize the gamma frequency throughout the discussion. Given that the power spectral density of the Local Field Potential (LFP) exhibits a log-log relationship between amplitude and frequency, as described by Buzsáki (2005) in "Rhythms of the Brain," and considering that the LFP is primarily generated through synaptic transmembrane currents (Buzsáki et al., 2012), it seems parsimonious to consider that the bulk of synaptic activity occurs at lower frequencies (e.g., theta). Since synaptic transmission represents the most direct form of inter-regional communication, one might wonder why gamma (characterized by lower amplitude rhythms) is esteemed so highly compared to the higher amplitude theta rhythm. Why isn't the theta rhythm, instead, regarded as the primary mode of communication across brain regions? A discussion exploring this question would be beneficial.

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